

AP EAMCET-2018-Engineering-22 Apr-Shift-2 Mathematics

1. If $f: \mathbb{R} \rightarrow [-1, 1]$ and $g: \mathbb{R} \rightarrow A$ are two subjective mappings and $\sin\left(g(x) - \frac{\pi}{3}\right) = \frac{f(x)}{2} \sqrt{4 - f^2(x)}$, then $A =$

(1) $\left[0, \frac{2\pi}{3}\right]$

(2) $[-1, 1]$

(3) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

(4) $(0, \pi)$

2. The domain of the function $f(x) = \sqrt{\frac{4 - x^2}{[x] + 2}}$, where $[x]$

denotes the greatest integer not more than x , is

(1) $(-\infty, -2) \cup (1, 2)$

(2) $(-\infty, -2) \cup (1, 2)$

(3) $(-\infty, -2) \cup [-1, 2]$

(4) $(-\infty, -1) \cup (1, 2)$

3. If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \dots n \text{ times}}}}$, then which one of the following is true?

(1) $a_n > 7 \forall n \geq 1$

(2) $a_n > 3 \forall n \geq 1$

(3) $a_n < 4 \forall n \geq 1$

(4) $a_n < 3 \forall n \geq 1$

4. If A is a square matrix of order 3, and $A^2 + A + 2I = 0$, then

(1) A can not be a skew-symmetric matrix

(2) $|A + I| = 0$

(3) A is non singular and $A^{-1} = (A + I)^{-1}$

(4) $|A||A + I| = 2$

5. If A is a square matrix of order 3, then consider the following statements

I: If $|A| = 0$ then $|\text{Adj } A| = 0$

II: If $|A| \neq 0$ then $|A^{-1}| = |A|^{-1}$

Which of the above statement is/are true?

- (1) Both I and II
- (2) Neither I nor II
- (3) I only
- (4) II only

6. The system of equations $x - 2y + 3z = 5$, $2x - 2y + z = 0$,
 $-x + 2y - 3z = 6$ has

- (1) infinitely many solutions
- (2) exactly two solutions
- (3) unique solution
- (4) no solution

7. The amplitude of $\sin \frac{\pi}{5} + i \left(1 - \cos \frac{\pi}{5} \right)$ is

- (1) $\frac{\pi}{15}$
- (2) $\frac{\pi}{10}$
- (3) $\frac{\pi}{5}$
- (4) $\frac{2\pi}{5}$

8. If a point P denotes a complex number $z = x + iy$ in the Argand plane and if $\frac{z+1}{z+i}$ is a purely real number, then the

locus of P is

- (1) $x + y + 1 = 0$
- (2) $x^2 + y^2 + x + y = 0$
- (3) $x^2 + y^2 + 2y + 1 = 0, (x, y) \neq (0, -1)$
- (4) $x + y + 1 = 0, (x, y) \neq (0, -1)$

9. If ω is a complex cube root of unity, then

$$\left[\frac{51 + 73\omega + 87\omega^2}{73 + 87\omega + 51\omega^2} + \frac{51 + 73\omega + 87\omega^2}{87 + 51\omega + 73\omega^2} \right]^{15} =$$

- (1) 1
- (2) -1
- (3) 0
- (4) 2

10. If $z \in \mathbb{C}$ and $iz^3 + 4z^2 - z + 4i = 0$, then a complex root of this equation having minimum magnitude is

- (1) $4i$

$$(2) \frac{1-i}{\sqrt{2}}$$

$$(3) \frac{\sqrt{3}+i}{2}$$

$$(4) \frac{1+i}{\sqrt{2}}$$

11. If α, β are the root of the equation $x^2 - 4x + 5 = 0$, then the quadratic equation whose roots are $\alpha^2 + \beta$ and $\alpha + \beta^2$ is

$$(1) x^2 + 10x + 34 = 0$$

$$(2) x^2 - 10x + 34 = 0$$

$$(3) x^2 - 10x - 34 = 0$$

$$(4) x^2 + 10x - 34 = 0$$

12. $f(x)$ is a quadratic expression such that $f(x)$ is negative

when $x \in \left(-\infty, -\frac{5}{3}\right) \cup (3, \infty)$ and positive when $x \in \left(-\frac{5}{3}, 3\right)$

$g(x)$ is another quadratic expression such that $g(x)$ is

negative when $x \in \left(3, -\frac{9}{2}\right)$ and positive when $x \in \left[-\frac{9}{2}, 3\right]$.

Then the sign of $f(x)g(x)$ in $[0, 5]$ is

- (1) Positive in $\left[0, \frac{9}{2}\right]$ and negative in $\left(\frac{9}{2}, 5\right)$
- (2) Positive in $[0, 3) \cup \left(3, \frac{9}{2}\right)$ and negative in $\left(\frac{9}{2}, 5\right]$
- (3) Positive in $[0, 3) \cup \left(3, \frac{9}{2}\right) \cup \left(\frac{9}{2}, 5\right]$
- (4) Negative in $[0, 3) \cup \left(3, \frac{9}{2}\right) \cup \left(\frac{9}{2}, 5\right]$

13. If $a, b, c \in \mathbb{R}$ be such that $4a + 2b + c > 0$ and $ax^2 + bx + c = 0$ has no real roots, then the value of $(c + a)(c + b)$ is

- (1) Greater than ab
- (2) Less than bc
- (3) Greater than ca
- (4) Less than $ab + bc + ca$

14. The minimum degree of a polynomial equation with rational coefficients having $\sqrt{3} + \sqrt{27}, \sqrt{2} + 5i$ as two of its roots is

- (1) 8
- (2) 6

(3) 4

(4) 2

15. If all the digits in the number 53426 are permuted in all possible ways and are arranged in decreasing order, then the number having rank 89, is

(1) 34265

(2) 43526

(3) 43526

(4) 43265

16. Three parallel straight lines L_1, L_2 and L_3 lie on the same plane. Consider 5 points on L_1 , 7 points on L_2 and 9 points on L_3 . Then the maximum possible number of triangles formed with vertices at these points, is

(1) 1330

(2) 1200

(3) 1201

(4) 129

17. If $a > 0$ and the coefficient of x^2 in the expression of

$$\left(ax^3 + \frac{c}{x}\right)^6 \text{ is } 60, \text{ then } ac^2 =$$

- (1) 2
- (2) 3
- (3) 4
- (4) 5

18. If $x = \frac{3}{4 \cdot 8} + \frac{3 \cdot 5}{4 \cdot 8 \cdot 12} + \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12 \cdot 16} + \dots$, then $2x^2 + 5x =$

- (1) $\frac{7}{8}$
- (2) 7
- (3) $\frac{7}{16}$
- (4) $\frac{7}{4}$

19. If $\frac{3x^2 + 1}{(x^2 + 1)(x^2 + 2)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 2} + \frac{Ex + F}{(x^2 + 2)^2}$, then

$$A + C + E =$$

- (1) 0

(2) $\frac{7}{3}$

(3) 1

(4) $\frac{4}{3}$

20. If $\tan\left(\frac{\pi}{4} + \frac{y}{2}\right) = \tan^3\left(\frac{\pi}{4} + \frac{x}{2}\right)$, then $\frac{3\sin x + \sin^3 x}{1 + 3\sin^2 x} =$

(1) 0

(2) 1

(3) $\sin 2y$

(4) $\sin y$

21.
$$\left[\frac{(\cos 252^\circ - \sin 126^\circ)(\cos 252^\circ + \sin 126^\circ)}{(\sin^2 126^\circ + \sin^2 186^\circ + \sin^2 66^\circ)} \right] =$$

(1) $\frac{3\sqrt{5}}{8}$

(2) $\frac{-3\sqrt{5}}{8}$

(3) $\frac{-3\sqrt{5}}{4}$

$$(4) \frac{3\sqrt{5}}{4}$$

22. If α, β, γ are any three angles, then

$$\cos \alpha + \cos \beta - \cos \gamma - \cos(\alpha + \beta + \gamma) =$$

$$(1) \quad 4 \cos \frac{\alpha + \beta}{2} \cos \frac{\beta + \gamma}{2} \cos \frac{\gamma + \alpha}{2}$$

$$(2) \quad 4 \cos \frac{\alpha + \beta}{2} \sin \frac{\beta + \gamma}{2} \sin \frac{\gamma + \alpha}{2}$$

$$(3) \quad 4 \cos \frac{\alpha + \beta}{2} \sin \frac{\beta - \gamma}{2} \sin \frac{\gamma - \alpha}{2}$$

$$(4) \quad 4 \sin \frac{\alpha + \beta}{2} \cos \frac{\beta + \gamma}{2} \cos \frac{\gamma + \alpha}{2}$$

23. The general solution of the equation

$$\sqrt{3 - 5 \sin x + \sin^2 x} + \cos x = 0 \text{ is}$$

$$(1) \quad n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

$$(2) \quad 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

$$(3) \quad (2n+1)\pi - \frac{\pi}{6}, n \in \mathbb{Z}$$

$$(4) \quad 2n\pi \pm \frac{5\pi}{6}, n \in \mathbb{Z}$$

24. Consider the following statements

$$\text{I.} \quad \sin^{-1}(y^2 - 4y + 6) + \cos^{-1}(y^2 - 4y + 6) = \frac{\pi}{2}, \forall y \in \mathbb{R}$$

$$\text{II.} \quad \sec^{-1}(y^2 - 4y + 6) + \operatorname{cosec}^{-1}(y^2 - 4y + 6) = \frac{\pi}{2}, \forall y \in \mathbb{R}$$

Which of the above statements is/are true?

- (1) only I
- (2) only II
- (3) Both I and II
- (4) Neither I nor II

25. If $\sec \theta \cosh y = \operatorname{cosec} x$ and $\operatorname{cosec} \theta \sinh y = \sec x$, then

$$\sinh^2 y =$$

- (1) $\cos^2 x$
- (2) $\cos x$
- (3) $\sin^2 x$
- (4) $\sin x$

26. Consider the following statements:

I. In triangle ABC , if $c = 6$ and $\cos C = \frac{-11}{25}$ then

$$R = \frac{25}{2\sqrt{14}}$$

II. In triangle ABC , if $a = 3, b = 4, c = 6$, then ABC is acute angled triangle.

Which of the above statements is/are true?

- (1) only I
- (2) only II
- (3) Both I and II
- (4) Neither I nor II

27. In triangle ABC , if $a = 3, b = 4, c = 6$, then

$$\frac{\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}}{\cot A + \cot B + \cot C} =$$

- (1) $\frac{13}{61}$
- (2) $\frac{169}{61}$
- (3) $\frac{61}{169}$

(4) $\frac{61}{13}$

28. If the reciprocals of the lengths of the sides of a ΔABC are in harmonic progression, then its exradii r_1, r_2, r_3 are in

- (1) Arithmetic progression
- (2) Geometric progression
- (3) Harmonic progression
- (4) Arithmetico-geometric progression

29. If P and Q are two points on the curve $y = 2^{x+2}$ such that $\overrightarrow{OP} \cdot \vec{i} = -1$ and $\overrightarrow{OQ} \cdot \vec{i} = 2$, then the magnitude of

$(\overrightarrow{OQ} - 4\overrightarrow{OP})$ is

- (1) 10
- (2) 1
- (3) 5
- (4) 100

30. P and Q are points on the straight line passing through the point $A(3\vec{i} + \vec{j} - \vec{k})$ and parallel to the vector $2\vec{i} - \vec{j} + 2\vec{k}$. If $AP = AQ = 3$, then the vector equation of the plane OPQ is

$$(1) \quad \vec{r} = (s + 5t)\vec{i} + 2s\vec{j} + (t - 3s)\vec{k}$$

$$(2) \quad \vec{r} = (3\vec{i} + \vec{j} - \vec{k}) + s(2\vec{i} - \vec{j} + 2\vec{k}) + t(5\vec{i} + \vec{k})$$

$$(3) \quad \vec{r} = (s + 5t)\vec{i} + 2s\vec{j} + (5s + t)\vec{k}$$

$$(4) \quad \vec{r} = (3t - s)\vec{i} + 2s\vec{j} + (t - 3s)\vec{k}$$

31. Let $\vec{m}$ be the unit vector orthogonal to the vector $\vec{i} - \vec{j} + \vec{k}$ and coplanar with the vectors $2\vec{i} + \vec{j}$ and $\vec{j} - \vec{k}$. If $\vec{a} = \vec{i} - \vec{k}$, then the length of the perpendicular from the origin to the plane $\vec{r} \cdot \vec{m} = \vec{a} \cdot \vec{m}$ is

$$(1) \quad \frac{1}{\sqrt{26}}$$

$$(2) \quad \frac{1}{\sqrt{5}}$$

$$(3) \quad \frac{5}{\sqrt{26}}$$

$$(4) \quad 1$$

32. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar unit vectors such that

$$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} + \vec{c}}{\sqrt{2}}, \text{ then the angle between } \vec{a} \text{ and } \vec{b} \text{ is}$$

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$

(4) $\frac{3\pi}{4}$

33. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{c} = (\vec{a} \times \vec{c}) + \vec{b}$, then the maximum value of $[\vec{a} \vec{b} \vec{c}]$ is

(1) 1

(2) $\frac{1}{2}$

(3) $\frac{3}{2}$

(4) 2

34. $\vec{\alpha}$, $\vec{\beta}$ and $\vec{\gamma}$ are non zero vectors such that $|\vec{\beta}| = |\vec{\gamma}| = 1$ and $|\vec{\alpha}| = 10$, then

$$(\vec{\alpha} \times (\vec{\beta} + \vec{\gamma})) \times (\vec{\beta} \times \vec{\gamma}) \cdot (\vec{\beta} - \vec{\gamma}) =$$

(1) 10

- (2) 1
- (3) 0
- (4) 12

35. The arithmetic mean and standard deviation of a data of nine numbers are 13 and 5 respectively. If 3 is included as the 10th item of the data, then the variance of the data of the ten number is

- (1) 23.5
- (2) 21.5
- (3) 31.5
- (4) 27

36. The variance of the following distribution is

Marks	1-3	3-5	5-7	7-9
Number of students	40	30	20	10

- (1) 2
- (2) 4
- (3) 6
- (4) 8

37. A and B are two events such that $P(A) = 0.58$, $P(B) = 0.32$ and $P(A \cap B) = 0.28$. Then the probability that neither A nor B occurs is

- (1) 0.38
- (2) 0.62
- (3) 0.72
- (4) 0.9

38. Two dice are thrown simultaneously. If A is event of getting the sum of the numbers on two dice as greater than or equal to 8 and B is the event of getting a number less than or equal to 3 on atleast one of the die. Then $P(B/A) =$

- (1) $\frac{5}{15}$
- (2) $\frac{6}{15}$
- (3) $\frac{7}{15}$
- (4) $\frac{8}{15}$

39. A bag contains 6 balls. If 4 balls are drawn at a time and all of them are found to be red, then the probability that exactly 5 of the balls in the bag are red is

(1) $\frac{10}{19}$

(2) $\frac{5}{21}$

(3) $\frac{1}{21}$

(4) $\frac{5}{7}$

40. If the probability distribution of a random variable X is given by

$$X = x_i \quad : \quad 0 \quad 1 \quad 2 \quad 3$$

$$P(X = x_i) : \quad \frac{1}{8} \quad \frac{3}{8} \quad 3K \quad K$$

Then the variance of X is

(1) 3

(2) $\frac{9}{4}$

(3) $\frac{3}{2}$

(4) $\frac{3}{4}$

41. A manufacturer of locks knows that 2% of his product is defective. If he sells the locks in boxes each with 100 locks and guarantees that not more than 2 locks will be defective in box, then the probability that a box will fail to meet the guaranteed quality is

(1) $1 - 5e^{-2}$

(2) $\sum_{k=2}^{100} {}^{100}C_k \left(\frac{1}{50}\right)^k \left(\frac{49}{50}\right)^{100-k}$

(3) 0.02

(4) $1 - 3e^{-2}$

42. The equation of the locus of a point $(2\cos\theta - 3, 3\sin\theta - 4)$ is

(1) $9x^2 + 4y^2 + 54x + 32y + 181 = 0$

(2) $4x^2 + 9y^2 + 54x + 32y + 109 = 0$

(3) $9x^2 + 4y^2 - 54x + 32y + 109 = 0$

(4) $9x^2 + 4y^2 + 54x + 32y + 109 = 0$

43. When the origin is shifted to the point $(2, 3)$ and then the coordinate axes are rotated through an angle $\frac{\pi}{3}$ in the counter clockwise sense, then the transformed equation of $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is

- (1) $3x^2 + 3y^2 - 1 = 0$
- (2) $(6 + \sqrt{3})x^2 - 2xy + (6 - \sqrt{3})y^2 - 2 = 0$
- (3) $4x^2 + 2y^2 - 1 = 0$
- (4) $(6 - \sqrt{3})x^2 + (6 + \sqrt{3})y^2 + 2xy = 0$

44. A straight line L with negative slope passes through the point $(1, 1)$ and cuts the positive coordinate axes at the points A and B . If O is the origin, then the minimum value of $OA + OB$ as L varies, is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

45. If the straight line $L \equiv 3x + 4y - k = 0$ cuts the line segment joining the points $P(2, -1)$ and $Q(1, 1)$ in the ratio 4:1, then the equation of the line parallel to the line $y = x$ and concurrent with the lines PQ and $L = 0$ is

- (1) $2x - 2y + 7 = 0$
- (2) $x - y + 1 = 0$
- (3) $5x - 5y - 3 = 0$
- (4) $y = x + 3$

46. The orthocenter and the centroid of $\triangle ABC$ are $(5, 8)$ and $\left(3, \frac{14}{3}\right)$ respectively. The equation of the side BC is $x - y = 0$. Given that the image of the orthocenter of a triangle with respect to any side lies on the circumcircle of that triangle, then the diameter of the circumcircle of $\triangle ABC$ is

- (1) $\sqrt{10}$
- (2) $2\sqrt{10}$
- (3) $4\sqrt{10}$
- (4) $8\sqrt{10}$

47. If a pair of perpendicular lines through the origin together with the straight line $2x + 3y = 6$ form an isosceles triangle then the area of that triangle (in sq. units) is

(1) $\frac{6}{\sqrt{13}}$

(2) $\frac{6}{13}$

(3) $\frac{36}{13}$

(4) $\frac{27}{13}$

48. If the equation $3x^2 + 7xy + 2y^2 + 2gx + 2fy + 2 = 0$ represents a pair of intersecting lines and the square of the distance of their point of intersection from the origin is $\frac{2}{5}$

then $f^2 + g^2 =$

(1) $\frac{25}{4}$

(2) 25

(3) 50

(4) $\frac{25}{2}$

49. From a point $P(0, b)$ two tangents are drawn to the circle $x^2 + y^2 = 16$ and these two tangents intersect x -axis at two points A and B . If the area of the triangle $\triangle PAB$ is minimum, then the equation of its circumcircle is

(1) $x^2 + y^2 = 16\sqrt{2}$

(2) $x^2 + y^2 = 64$

(3) $x^2 + y^2 = 32$

(4) $x^2 + y^2 = 4\sqrt{2}$

50. If the angle between the tangents drawn to the circle $x^2 + y^2 - 12x - 16y = 0$ at the points where the line $5y = 5x + k$ cut the circle is 60° , then the value of k is,

(1) $5 + \sqrt{2}$

(2) $5(2 \pm 5\sqrt{2})$

(3) $2 \pm 5\sqrt{2}$

(4) $5 \pm 5\sqrt{2}$

51. If a circle S with radius 5 touches the circle $x^2 + y^2 - 6x - 4y - 12 = 0$ at $(-1, -1)$ then the length of the tangent from the centre of the circle S to the given circle is

- (1) $5\sqrt{3}$
- (2) $\sqrt{65}$
- (3) 10
- (4) $3\sqrt{11}$

52. If a circle S passing through the point $(3, 4)$ cuts the circle $x^2 + y^2 = 36$ orthogonally, then the locus of the centre of S is

- (1) $x^2 + y^2 - 6x - 8y + 11 = 0$
- (2) $6x + 8y - 61 = 0$
- (3) $x^2 + y^2 - 8x - 6y + 11 = 0$
- (4) $6x + 8y + 11 = 0$

53. The line $x - 2 = 0$ cuts the circle $x^2 + y^2 - 8x - 2y + 8 = 0$ at A and B . The equation of the circle passing through the points A and B and having least radius is

- (1) $x^2 + y^2 - 4x + 2y - 1 = 0$

(2) $x^2 + y^2 - 4x - 2y = 0$

(3) $x^2 + y^2 - 4x - 2y + 1 = 0$

(4) $x^2 + y^2 - 4x + 4y = 0$

54. If a perpendicular drawn through the vertex O of the parabola $y^2 = 4ax$ to any of its tangent meets the tangent at N and the parabola at M, then $ON \cdot OM =$

(1) $4a^2$

(2) $3a^2$

(3) $2a^2$

(4) a^2

55. Let α_1 and α_2 be the ordinates of two points A and B on a parabola $y^2 = 4ax$ and let α_3 be the ordinate of the point of intersection of its tangents at A and B. Then $\alpha_3 - \alpha_2 =$

(1) $\alpha_3 - \alpha_1$

(2) $\alpha_3 + \alpha_1$

(3) α_1

(4) $\alpha_1 - \alpha_3$

56. The equations of the latus recta of the ellipse $9x^2 + 4y^2 - 18x - 8y - 23 = 0$ are

(1) $x = -1 \pm \sqrt{5}$

(2) $y = 1 \pm \sqrt{5}$

(3) $x = 1 \pm \frac{2\sqrt{5}}{3}$

(4) $y = 2 \pm \sqrt{5}$

57. The equation of the tangent to the ellipse $4x^2 + 9y^2 = 36$ at the end of the latus rectum lying in the second quadrant, is

(1) $\sqrt{5}x - 3y + 1 = 0$

(2) $x - 3y + \sqrt{5} = 0$

(3) $\sqrt{5}x - 3y + 3 = 0$

(4) $\sqrt{5}x - 3y + 9 = 0$

58. If the product of the lengths of the perpendiculars from any point on the hyperbola $16x^2 - 25y^2 = 400$ to its asymptotes is p and the angle between the two asymptotes is θ then

$$p \tan \frac{\theta}{2} =$$

(1) $\frac{400}{41}$

(2) $\frac{320}{41}$

(3) $\frac{4}{5}$

(4) $\frac{25}{16}$

59. $A(3, 2, -1)$, $B(4, 1, 1)$, $C(6, 2, 5)$ and $D(3, 3, 3)$ are four points G_1, G_2, G_3 and G_4 respectively are the centroids of the triangles BCD , CDA , DAB and ABC . The point of concurrence of the lines AG_1, BG_2, CG_3 and DG_4 is

(1) $(4, 2, 2)$

(2) $(2, 4, 2)$

(3) $(2, 2, 4)$

(4) $(2, 2, 2)$

60. The acute angle between the lines whose direction cosines are given by the equations $l + m + n = 0$ and $2lm + 2ln - mn = 0$ is

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$

(4) $\frac{2\pi}{5}$

61. A variable plane passes through a fixed point (α, β, γ) and meets the coordinate axes in A, B, and C. Let P_1 , P_2 and P_3 be the planes passing through A, B, C and parallel to the coordinate planes YZ, ZX, XY respectively. Then the locus of the point of intersection of the planes P_1 , P_2 and P_3 is

(1) $\alpha x + \beta y + \gamma z = 1$

(2) $\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 1$

(3) $\alpha x^2 + \beta y^2 + \gamma z^2 = 1$

(4) $\alpha\beta x + \beta\gamma y + \alpha\gamma z = 1$

62. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \cos 2x}{\cot 3x (3^{\sin 2x} - 1)} =$

$$(1) \frac{1}{3\log 9}$$

$$(2) \frac{2}{3\log 3}$$

$$(3) \frac{1}{3\log 3}$$

$$(4) \frac{3}{\log 3}$$

$$63. \lim_{n \rightarrow \infty} n^{-nk} \left\{ (n+1) \left(n + \frac{1}{2} \right) \left(n + \frac{1}{2^2} \right) \dots \left(n + \frac{1}{2^{k-1}} \right) \right\}^n =$$

$$(1) 2$$

$$(2) e^{2\left(1 - \frac{1}{2^k}\right)}$$

$$(3) 2\left(1 - \frac{1}{2^k}\right)$$

$$(4) e^2$$

64. If a and b ($a > b$) are points of discontinuity of the function

$$f(x) = \begin{cases} 3 - 2x & , \text{ for } x \leq 0 \\ 2x + 3 & , \text{ for } 0 < x \leq 1 \\ 2x^2 - 3x & , \text{ for } 1 < x < 2 \\ 2x - 3 & , \text{ for } 2 \leq x < 3 \\ |x| & , \text{ for } x \geq 3 \end{cases}$$

then $3a - b =$

- (1) 3
- (2) 7
- (3) 5
- (4) 1

65. For $-1 < x < 1$, if $f(x) = \cos^2 \left(\tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$ then $f'(x) =$

- (1) $\frac{1}{2}$
- (2) 1
- (3) -1
- (4) $-\frac{1}{2}$

66. $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(0) = 1$ and for all $x, y \in \mathbb{R}$

$f(xy+1) = f(x)f(y) - f(y) - x + 2$ then $\frac{df}{dx}$ at $x = e$ is

- (1) 0
- (2) -1
- (3) e
- (4) 1

67. $y = \sin(\log(x^2 + 2x + 1)) \Rightarrow (x+1)^2 \frac{d^2y}{dx^2} + (x+1) \frac{dy}{dx} =$

- (1) y
- (2) $-4y$
- (3) $4y$
- (4) $-y$

68. The acute angle between the tangents drawn at the point at the point of intersection (other than the origin) of the curves $x^2 = 4y$ and $y^2 = 4x$ is

- (1) $\tan^{-1}\left(\frac{1}{2}\right)$
- (2) $\sin^{-1}\left(\frac{3}{5}\right)$

(3) $\cos^{-1}\left(\frac{1}{3}\right)$

(4) $\tan^{-1}\left(\frac{2}{3}\right)$

69. If $x > 0$ then $\frac{x}{1+x} - \log(1+x)$

- (1) is less than zero
- (2) is greater than zero
- (3) is equal to zero
- (4) takes all the real values

70. On the curve $y = x^3$, the point at which the tangent line is parallel to the chord joining the points $(-1, -1)$ and $(2, 8)$, is

- (1) $(1, -1)$
- (2) $(2, -8)$
- (3) $(1, 1)$
- (4) $(3, 27)$

71. If the petrol burnt in driving a motor boat varies as the cube of the velocity, then the speed (in km/hour) of the boat going against a water flow of C kms/hour so that the quantity of petrol burnt is minimum is

(1) $\frac{2C}{3}$

(2) $\frac{3C}{2}$

(3) $\frac{4C}{3}$

(4) $\frac{3C}{4}$

72. For $x < 1$, $\int \frac{x - x^2}{\sqrt{1-x}} dx =$

(1) $\frac{4}{35}(1-x)^{\frac{3}{2}} - \frac{2}{5}(1-x)^{\frac{5}{2}} - 2\sqrt{1-x} + c$

(2) $\frac{4}{3}(1-x)^{\frac{3}{2}} - \frac{2}{3}(1-x)^{\frac{5}{2}} - 2\sqrt{1-x} + c$

(3) $\frac{2}{3}(1-x)^{\frac{3}{2}} - 2\sqrt{1-x} + c$

(4) $-\frac{2}{15}(1-x)^{\frac{3}{2}}(2+3x) + c$

$$73. \int \frac{dx}{(2 \sin x + \sec x)^4} = A(1 + \tan x)^{-5} + B(1 + \tan x)^{-6} + C(1 + \tan x)^{-7} + k$$

, then $A + B + C =$

(1) $\frac{-86}{105}$

(2) $\frac{-1}{105}$

(3) $\frac{-26}{105}$

(4) $\frac{-16}{105}$

$$74. \int \frac{2x^2 - 1 + x^2 \sqrt{x^2 + 4}}{x^2(x^2 + 4)} dx =$$

(1) $\frac{9}{8} \tan^{-1} \frac{x}{2} + \frac{1}{4x} + \cosh^{-1} \frac{x}{2} + c$

(2) $\frac{9}{8} \tanh^{-1} \frac{x}{2} + \frac{1}{4x} + \sinh^{-1} \frac{x}{2} + c$

(3) $\frac{9}{16} \log \left| \frac{x+2}{x-2} \right| + \frac{1}{4x} + \log \left| \frac{x + \sqrt{x^2 + 4}}{2} \right| + c$

$$(4) \quad \frac{9}{16} \log \left| \frac{2-x}{2+x} \right| + \frac{1}{4x} + \cosh^{-1} \frac{x}{2} + c$$

75. For $n \geq 2$, let $I_n = \int_0^{\frac{\pi}{4}} \tan^m x dx$ and $F_n = I_n + I_{n-2}$. Then

$$F_n - F_{n+1} =$$

$$(1) \quad \frac{1}{n}$$

$$(2) \quad \frac{1}{n-1}$$

$$(3) \quad \frac{1}{n(n-1)}$$

$$(4) \quad 1+n$$

$$76. \quad \lim_{n \rightarrow \infty} n \left[\frac{1}{3n^2 + 8n + 4} + \frac{1}{3n^2 + 16n + 16} + \dots + \frac{1}{15n^2} \right] =$$

$$(1) \quad \frac{1}{2} \log \frac{9}{5}$$

$$(2) \quad \frac{1}{4} \log \frac{9}{5}$$

$$(3) \quad 2 \log \frac{9}{5}$$

(4) $\frac{1}{4} \log \frac{5}{9}$

77. $\int |x^2 - 3x + 2| dx =$

(1) $\frac{3}{2}$

(2) $\frac{1}{6}$

(3) $\frac{11}{6}$

(4) $\frac{11}{2}$

78. OABC is a unit square where O is the origin and $B = (1, 1)$.

The curves $y^2 = x$ and $x^2 = y$ divide the area of the square into three parts OABO, OBO, and OBCO. If a_1, a_2, a_3 are the areas (in sq. units) of these parts respectively, then $a_1 + 2a_2 + 3a_3 =$

(1) 1

(2) 2

(3) 6

(4) 64

79. The differential equation corresponding to the family of parabolas $y^2 = 4a(x + a)$, where a is the parameter, is

(1) $y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} + y = 0$

(2) $y\left(\frac{dy}{dx}\right)^2 - 2x\frac{dy}{dx} - y = 0$

(3) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(4) $y = 2x\frac{dy}{dx}$

80. The general solution; of $\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$

(1) $y - x^2 = c \sec x$

(2) $y \cos x = x^2 \sec x + c$

(3) $y \sec x = x^2 + c \cos x$

(4) $y = x^2 + c \cos x$

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Physics

81. In a system, unit of mass is A kg, length is B m and time is C s, then the value of 10 N in this system is
- (1) $10 A^{-1} B^{-1} C^{-2}$
 - (2) $10 A^{-1} B^{-1} C^2$
 - (3) $10 ABC^{-2}$

(4) $5 A^{-1}BC^2$

82. Assertion (A) : The angle between acceleration and velocity of a body in one dimensional motion is always zero.

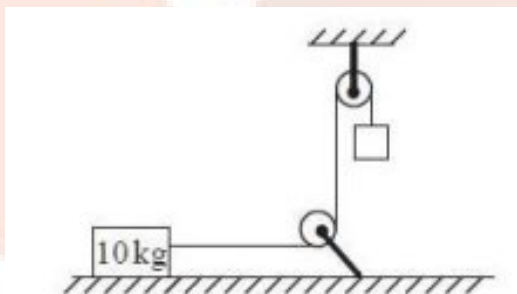
Reason (R) : One dimensional motion is along a straight line.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

83. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j}) \text{ ms}^{-1}$. The equation of its path is $(g = 10 \text{ ms}^{-1})$

- (1) $y = 2x - 5x^2$
- (2) $y = x - 5x^2$
- (3) $4y = 2x - 5x^2$
- (4) $y = 2x - 25x^2$

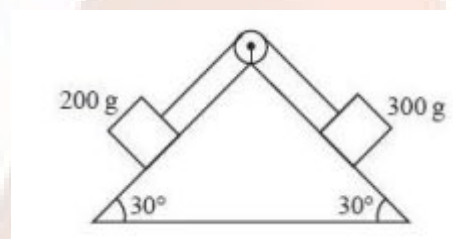
84. A body projected with some velocity at an angle 45° with the horizontal from the origin in X-Y plane passes through a point at (4, 3) m. Its horizontal range is
- (1) 10 m
 - (2) 14 m
 - (3) 18 m
 - (4) 16 m
85. A block of mass 10 kg is placed on a horizontal frictionless surface and is attached to a cord which passes over two light frictionless pulleys as shown in the figure. The hanging block tied to the other end of the cord is initially at rest 2 m above the horizontal floor. If the hanging block strikes the 2 s after the system is released, the weight of the hanging block is ____ ($g = 10 \text{ ms}^{-2}$)



- (1) 22.22 N
- (2) 11.11 N
- (3) 1.11 N

(4) 2.22 N

86. A double inclined plane as shown in figure has fixed horizontal base and smooth faces with the same angle of inclination of 30° . A block of mass 300 g is on one face and is connected by a cord passing over a frictionless pulley to a second block of mass 200 g kept on another face. The acceleration with which the system of the blocks moves is _____% of acceleration due to gravity.



- (1) 5
(2) 10
(3) 15
(4) 20
87. A canon shell fired breaks into two equal parts at its highest point. If one part retraces the path to the canon with kinetic energy E_1 and kinetic energy of the second part is E_2 then
- (1) $E_2 = 15E_1$

- (2) $E_2 = E_1$
- (3) $E_2 = 4E_1$
- (4) $E_2 = 9E_1$

88. A uniform chain of mass 'm' and length 'l' is on a smooth horizontal table with $\left(\frac{1}{n}\right)^{\text{th}}$ part of its length is hanging from one end of the table. The velocity of the chain when it completely slips off the table is

- (1) $\sqrt{gl\left(1 - \frac{1}{n^2}\right)}$
- (2) $\sqrt{2gl\left(1 + \frac{1}{n^2}\right)}$
- (3) $\sqrt{2gl\left(1 - \frac{1}{n^2}\right)}$
- (4) $\sqrt{2gl}$

89. Two particles of masses in the ratio 1 : 2 are placed along a vertical line. The lighter particle is raised through a height of 9 cm. To raise the centre of mass of the system by 2 cm, the heavier particle should be _____

- (1) moved 1.5 cm downward
- (2) moved 2 cm upward
- (3) moved 1.5 cm upward
- (4) moved 2 cm downward

90. A solid sphere and a ring of same radius roll down an inclined plane without slipping. Both start from rest from the top of the inclined plane. If the sphere and the ring reach the bottom of the inclined plane with velocities v_s and v_r respectively, then $\frac{v_r^2}{v_s^2}$ is

- (1) 0.2
- (2) 0.5
- (3) 0.7
- (4) 0.9

91. A particle is executing S.H.M. The time taken for $\left(\frac{3}{8}\right)^{\text{th}}$ of oscillation from extreme position is 'X'. Then the time taken for the particle to complete $\left(\frac{5}{8}\right)^{\text{th}}$ of oscillation from mean positions is

(1) $\frac{5X}{4}$

(2) $\frac{7X}{4}$

(3) $\frac{21X}{8}$

(4) $\frac{7X}{12}$

92. An object is thrown vertically upward from the surface of the earth with a velocity x times the escape velocity on the earth ($x < 1$), the maximum height to which it raises from the centre of the earth is

(2) (radius of earth is R)

(1) $R(1 - x^2)$

(2) $\frac{R}{(1 - x^2)}$

(3) $\frac{(1 - x^2)}{R}$

(4) $\frac{x^2}{1 - R}$

93. A sphere of mass 2 kg and diameter 4.5 cm is attached to the lower end of steel wire of 2 m length and area of cross section $0.24 \times 10^{-6} \text{ m}^2$. The wire is suspended from 205 cm high ceiling of a room. When the system is made to oscillate as a simple pendulum, the sphere just grazes the floor at its lowest position. The velocity of the sphere at the lowest position is

(3) (Young's modulus of steel = $2 \times 10^{11} \text{ Nm}^{-2}$ and acceleration due to gravity = 10 ms^{-2})

(1) 10 ms^{-1}

(2) 12 ms^{-1}

(3) 15 ms^{-1}

(4) 18 ms^{-1}

94. A spherical body of density ρ is floating half immersed in a liquid of density d . If σ is the surface tension of the liquid, then the diameter of the body is

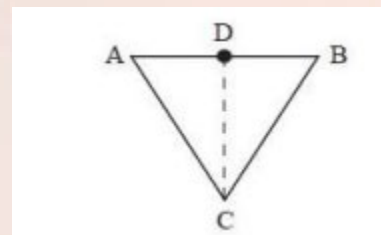
(1) $\sqrt{\frac{3\sigma}{g(2\rho - d)}}$

(2) $\sqrt{\frac{6\sigma}{g(2\rho - d)}}$

$$(3) \sqrt{\frac{4\sigma}{g(2\rho - d)}}$$

$$(4) \sqrt{\frac{12\sigma}{g(2\rho - d)}}$$

95. As shown in the figure, an equilateral triangle ABC is formed by joining three rods of equal length and D is the midpoint of AB. Coefficient of linear expansion of the material of AB is α_1 and that of AC and BC is α_2 . If the length DC remains constant for small changes in temperature, then



(4)

- (1) $\alpha_1 = \alpha_2$
 (2) $\alpha_1 = 4\alpha_2$
 (3) $\alpha_2 = 4\alpha_1$
 (4) $\alpha_1 = \frac{\alpha_2}{2}$

96. Match the following List-I with List-II.

List-I

List-II

- | | |
|-----------------------------------|---|
| (A) When ice melt into water | (I) Volume increases |
| (B) When water changes into steam | (II) Volume decreases |
| (C) Melting point of ice | (III) Increases with increase of pressure |
| (D) Boiling point of water | (IV) Decreases with increase of pressure |

The correct answer is

- | | | | | |
|-----|-----|----|-----|-----|
| (1) | A | B | C | D |
| | I | II | III | IV |
| (2) | A | B | C | D |
| | II | I | IV | III |
| (3) | A | B | C | D |
| | III | II | IV | I |
| (4) | A | B | C | D |
| | II | I | III | IV |

97. A cylindrical vessel of uniform cross section consisting of a gas of $\gamma = 1.5$ is divided into two part A and B using a piston. Initially the piston is kept fixed such that part A has pressure P and volume $5V$ and the part B has pressure $8P$ and volume V . If the piston is let free and the gas is allowed to undergo adiabatic process, the final volume of the gas in part A is _____

- (1) $3V$
- (2) $\frac{8}{3}V$
- (3) $\frac{10}{3}V$
- (4) $\frac{13}{3}V$

98. A diatomic ideal gas is used in a Carnot's engine as working substance. During adiabatic expansion of the cycle, if the volume of the gas increases from V to $32V$ then the efficiency of the engine is

- (1) 0.25
- (2) 0.5
- (3) 0.67
- (4) 0.75

99. The absolute temperature at which the rms speed of a hydrogen molecule is equal to its escape speed from the Moon's surface is
(5) _____ (radius of Moon is R , g is acceleration due to gravity on Moon's surface, m is the mass of hydrogen molecule and k is Boltzmann constant)

(1) $\frac{mgR}{2k}$

(2) $\frac{2mgR}{k}$

(3) $\frac{3mgR}{2k}$

(4) $\frac{2mgR}{3k}$

100. An object of density 2000 kgm^{-3} is hung from a thin light wire. The fundamental frequency of the transverse waves in the wire is 200 Hz. If the object is immersed in water such that half of its volume is submerged, then the fundamental frequency of the transverse waves in the wire is _____

(1) 200.0 Hz

(2) 173.2 Hz

(3) 100.0 Hz

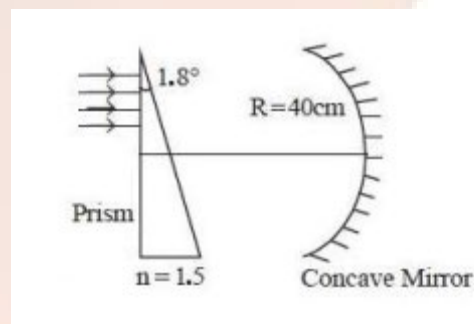
(4) 144.4 Hz

101. An observer and a source emitting sound of frequency 120 Hz are on the X-axis. The observer is stationary while the source of sound is in motion given by the equation $x = 3 \sin \omega t$ (x in meters and t in second). If the difference between the maximum

and minimum frequencies of the sound observed by the observer is 22 Hz, then the value of ω is (Speed of sound in air = 330 ms^{-1})

- (1) 33 rads^{-1}
- (2) 36 rads^{-1}
- (3) 20 rads^{-1}
- (4) 10 rads^{-1}

102. As shown in the figure, a parallel beam of light incident on the upper part of a prism of angle 1.8° and material of refractive index 1.5. The light emerging out from the prism falls on a concave mirror of radius of curvature 40 cm. The distance of the point from the principle axis of the mirror where the light rays are focussed after reflection from the mirror is



- (6)
- (1) 4.76 cm
- (2) 1.57 cm
- (3) 3.14 cm

(4) 6.28 cm

103. A microscope has an objective of aperture 8 mm and focal length of 5 cm. The minimum separation between two object to be just resolved by the microscope is

(7) (Wavelength of light used = $5500 \text{ \AA}$)

(1) $2.2 \text{ }\mu\text{m}$

(2) $3.4 \text{ }\mu\text{m}$

(3) $4.2 \text{ }\mu\text{m}$

(4) $3.6 \text{ }\mu\text{m}$

104. The electric field due to short electric dipole at a distance 'r' on the axial line from its mid point is x times the electric field at a distance 2r on the equatorial line from the mid-point of dipole.

Then the value of x is

(1) 16

(2) 9

(3) 25

(4) 36

105. A point charge 'q' is placed at origin. Let $\vec{E}_A$, $\vec{E}_B$ and $\vec{E}_C$ be the electric fields at three points A (1, 2, 3), B(1, 1, -1) and (2, 2, 2) respectively due to the charge 'q'. Then the relation between them is

(a) $\vec{E}_A \perp \vec{E}_B$

(b) $\vec{E}_A \parallel \vec{E}_C$

(c) $|\vec{E}_B| = 4|\vec{E}_C|$

(d) $|\vec{E}_B| = 8|\vec{E}_C|$

(1) a, d are correct

(2) b, d are correct

(3) a, c are correct

(4) b, c are correct

106. An electric dipole consists of two particles each of mass 1 kg separated by 1 m carrying charges $1 \mu\text{C}$ and $-1 \mu\text{C}$ respectively. It is equilibrium in a uniform electric field of $2 \times 10^4 \text{ Vm}^{-1}$. If it is deflected by a small angle 2° , minimum time taken by it to come back again to the mean position is (in seconds)

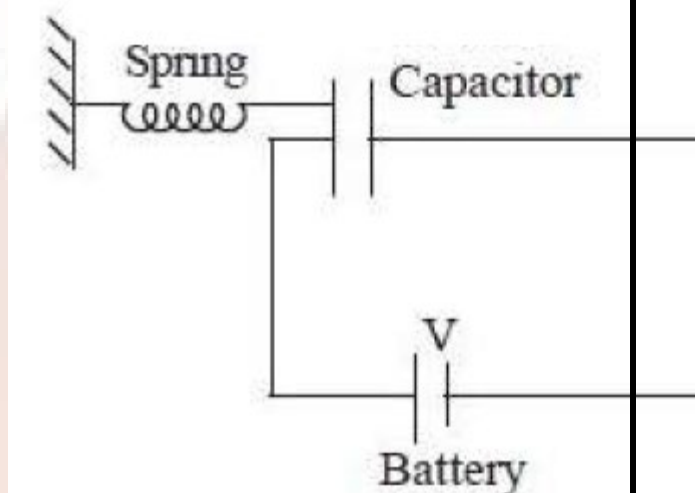
(1) 2.5π

(2) 2π

(3) 5π

(4) 4π

107. One plate of a parallel plate capacitor is connected to a spring as shown in the figure. The area of each plate of the capacitor is A and the distance between the plates is d when the battery is not connected and the spring is unstretched. After connecting the battery, in the steady state, the distance between the plates is $0.74d$, then the force constant of the spring is



(8)

(1) $\frac{3\epsilon_0 V^2 A}{8 d^3}$

(2) $\frac{8\epsilon_0 V^2 A}{3 d^3}$

(3) $\frac{9\epsilon_0 V^2 A}{32 d^3}$

$$(4) \quad \frac{32 \epsilon_0 V^2 A}{9 d^3}$$

108. Two cell P and Q each of emf 2.16 V are connected in series with a resistor of $19.6 \, \Omega$. An ideal voltmeter reads 2 V when connected across the cell P and 1.92 V when connected across the cell Q. The ratio of the internal resistance of the cells P and Q is

- (1) 1 : 2
- (2) 2 : 3
- (3) 3 : 4
- (4) 1 : 3

109. A resistor has bands with colours orange, green, silver and gold. Then the resistance of the resistor is

- (1) $(350 \pm 5) \, \text{m}\Omega$
- (2) $(350 \pm 17.5) \, \text{m}\Omega$
- (3) $(35 \pm 5\%) \, \text{m}\Omega$
- (4) $(250 \pm 5\%) \, \text{m}\Omega$

110. A beam of protons enters a uniform magnetic field of 0.314T with a velocity $4 \times 10^5 \text{ ms}^{-1}$ in a direction making an angle 60° with the direction of the magnetic field. The path of the beam is (mass of proton = $1.6 \times 10^{-27} \text{ kg}$)
- (1) a circle of radius 0.2 m
 - (2) a straight line
 - (3) a helix with a pitch 4 cm
 - (4) a helix with a pitch 4 mm
111. The magnetic field due to a current carrying loop of radius 3 cm at a point on its axis at a distance of 4 cm from its centre is $54 \mu\text{T}$. Then the value of magnetic field at the centre of the loop is _____
- (1) $250 \mu\text{T}$
 - (2) $150 \mu\text{T}$
 - (3) $75 \mu\text{T}$
 - (4) $125 \mu\text{T}$
112. A short bar magnet of magnetic moment 0.21 Am^2 is placed with its axis perpendicular to the direction of the horizontal components of the Earth's magnetic field. The distance of the

point on the axis of the magnet from the centre of the magnet where the resultant magnetic field is inclined at 45° with the horizontal component of the Earth's field direction is _____

(9) _____ (Horizontal component of the Earth's magnetic field = 4.2×10^{-5} T)

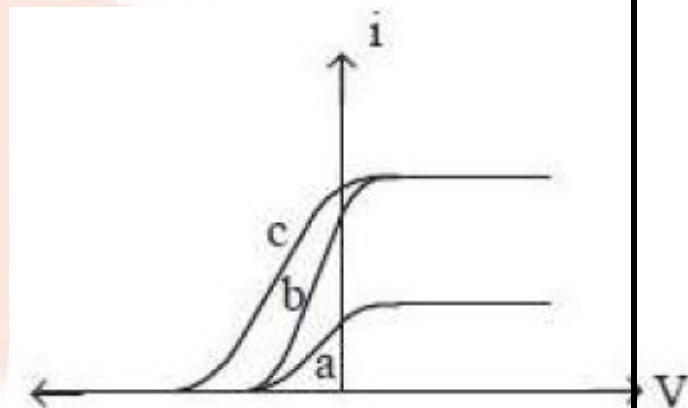
- (1) 12 cm
- (2) 20 cm
- (3) 5 cm
- (4) 10 cm

113. The length of a wire required to make a solenoid of length ' l ' and self induction ' L ' is

- (1) $\sqrt{\frac{4\pi Ll}{\mu_0}}$
- (2) $\sqrt{\frac{Ll}{4\pi\mu_0}}$
- (3) $\sqrt{\frac{2\pi Ll}{\mu_0}}$
- (4) $\sqrt{\frac{\mu_0 Ll}{4\pi}}$

114. An inductor and a resistor are connected in series to an ac source. The current in the circuit is 500 mA if the applied ac voltage is $8\sqrt{2}$ V at a frequency of $\frac{175}{\pi}$ Hz and the current in the circuit is 400 mA if the same ac voltage at a frequency of $\frac{225}{\pi}$ Hz is applied. The values of the inductance and resistance are respectively
- (1) 60 mH, 71 Ω
 - (2) $\sqrt{60}$ mH, 71 Ω
 - (3) $\sqrt{60}$ mH, $\sqrt{71}$ Ω
 - (4) 60 mH, $\sqrt{71}$ Ω
115. An electromagnetic waves of frequency 2 MHz propagates from vacuum to a non magnetic medium of relative permittivity 9. Then its wavelength
- (1) Increases by 100 m
 - (2) Increases by 50 m
 - (3) Decreases by 50 m
 - (4) Decreases by 100 m

116. The figure shows the variation of photocurrent (i) with anode potential (V) for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a, b and c respectively. Then



(10)

(1) $f_a = f_b$ and $I_a \neq I_b$

(2) $f_a = f_c$ and $I_a = I_c$

(3) $f_a = f_b$ and $I_a = I_b$

(4) $f_b = f_c$ and $I_b = I_c$

117. A stationary hydrogen atom undergoes a transition from $n = 5$ to $n = 4$. Recoil speed of the atom is (R = Rydberg constant, h = Plank's constant, m = mass of the proton)

(1) $\frac{Rh}{m}$

(2) $\frac{9m}{400Rh}$

(3) $\frac{9Rh}{400m}$

(4) $\frac{7Rh}{400}$

118. The half life of ${}_{92}^{238}\text{U}$ against α -decay is 13.86×10^{16} s. The activity of 1 g sample of ${}_{92}^{238}\text{U}$ is _____

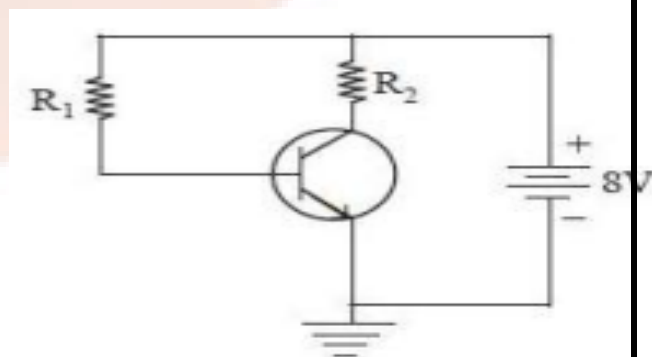
(1) $1.26 \times 10^4 \text{ s}^{-1}$

(2) $1.26 \times 10^{-4} \text{ s}^{-1}$

(3) $12.6 \times 10^4 \text{ s}^{-1}$

(4) $12.6 \times 10^{-4} \text{ s}^{-1}$

119. An npn transistor is connected in common emitter configuration as shown in the figure. If the collector current is 5 mA, $V_{BE} = 0.6 \text{ V}$, $V_{CE} = 3 \text{ V}$ and common emitter current amplification factor is 50, then the values of R_1 and R_2 are respectively



(11)

(1) $1 \text{ k}\Omega, 74 \text{ k}\Omega$

- (2) $74 \text{ k}\Omega$, $1 \text{ k}\Omega$
- (3) $37 \text{ k}\Omega$, $2 \text{ k}\Omega$
- (4) $2 \text{ k}\Omega$, $37 \text{ k}\Omega$

120. The maximum distance between the transmitting and receiving TV towers is 65 km. If the ratio of the heights of the transmitting and receiving towers respectively are (Radius of earth 6400 km)

- (1) 51.2 m, 80 m
- (2) 70.3 m, 95.7 m
- (3) 30 m, 65 m
- (4) 25 m, 75 m

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121. The wavelength corresponding to electronic transition between two orbits of hydrogen atom is $912 \text{ \AA}$. The wavelength (in $\text{\AA}$) for the same electronic transition in Li^{2+} is
- (1) 101.3
 - (2) 202.6
 - (3) 303.9
 - (4) 50.65
122. The ratio of lowest energy in terms of wave numbers of Balmer and Lyman series of lines of atomic spectrum of hydrogen is
- (1) 5:27
 - (2) 27:5
 - (3) 20:27
 - (4) 27:2

123. Which of the following represents the correct order of ionic radii?

- (1) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{O}^{2-} > \text{F}^-$
- (2) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
- (3) $\text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ >$
- (4) $\text{O}^{2-} > \text{F}^- > \text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$

124. The species, which has the bond order same as that of F_2 molecule is

- (1) O_2^+
- (2) O_2^{2-}
- (3) O_2
- (4) N_2^+

125. The hybridization of an atom 'X' with atomic number 27 in $[\text{XF}_6]^{3-}$ is

- (1) dsp^2
- (2) d^2sp^3
- (3) sp^3d^2
- (4) sp^3

126. The temperature of 4.0 moles of a gas occupying 5dm^3 at 3.32 bar is

$$(R = 0.083 \text{ bar dm}^3 \text{ K}^{-1} \text{ mol}^{-1})$$

- (1) 25 K
- (2) 50 K
- (3) 75 K
- (4) 100 K

127. To 50mL of 0.1 N Na_2CO_3 solution 150 mL of water is added.

What is the molarity of resultant solution?

- (1) $\frac{M}{40}$
- (2) $\frac{M}{20}$
- (3) $\frac{M}{80}$
- (4) $\frac{M}{30}$

128. Match the following

List-I	List-II
--------	---------

A) At constant volume the change in internal energy of a system	I) $W = -2.303nRT \log \frac{V_f}{V_i}$
B) Isothermal irreversible change	II) $W_{\text{adia}} = \Delta U$
C) Isothermal reversible change	III) $q_v = \Delta U$
D) Adiabatic change	IV) $W = -P_{\text{ex}} (V_f - V_i)$
	V) $\Delta U = \Delta H - nRT$

The correct answer is

- (1) $\frac{\underline{A}}{\underline{V}}$ $\frac{\underline{B}}{\underline{\text{III}}}$ $\frac{\underline{C}}{\underline{\text{IV}}}$ $\frac{\underline{D}}{\underline{\text{I}}}$
- (2) $\frac{\underline{A}}{\underline{\text{IV}}}$ $\frac{\underline{B}}{\underline{\text{I}}}$ $\frac{\underline{C}}{\underline{\text{III}}}$ $\frac{\underline{D}}{\underline{\text{V}}}$
- (3) $\frac{\underline{A}}{\underline{\text{III}}}$ $\frac{\underline{B}}{\underline{\text{IV}}}$ $\frac{\underline{C}}{\underline{\text{I}}}$ $\frac{\underline{D}}{\underline{\text{II}}}$
- (4) $\frac{\underline{A}}{\underline{\text{III}}}$ $\frac{\underline{B}}{\underline{\text{V}}}$ $\frac{\underline{C}}{\underline{\text{I}}}$ $\frac{\underline{D}}{\underline{\text{II}}}$

129. The pH of a buffer solution formed by mixing 30 mL of 0.1 M NH_4OH and 30 mL of 1 M NH_4Cl solution is 8.6. The pK_b of NH_4OH is

- (1) 5.4
- (2) 4.4
- (3) 5.6
- (4) 4.2

130. The solubility products of three sparingly soluble salts AB, A_2B and AB_3 are respectively 4.0×10^{-20} , 3.2×10^{-11} and 2.7×10^{-31} . The increasing order of their solubility is

- (1) $\text{AB} < \text{AB}_3 < \text{A}_2\text{B}$
- (2) $\text{AB}_3 < \text{AB} < \text{A}_2\text{B}$
- (3) $\text{A}_2\text{B} < \text{AB}_3 < \text{AB}$
- (4) $\text{A}_2\text{B} < \text{AB} < \text{AB}_3$

131. Identify the correct statements from the following

- a) Zn reacts with dilute HCl and aqueous NaOH solution separately and liberates hydrogen
- b) Ti and Zr form interstitial hydrides

c) The viscosity of H_2O is more than the viscosity of D_2O

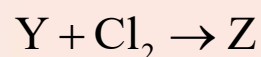
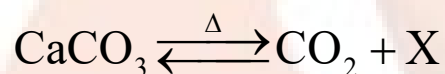
(1) a, b, c

(2) a, c

(3) a, b

(4) b, c

132. What are X, Y and Z in the following reactions?



	<u>X</u>	<u>Y</u>	<u>Z</u>
(1)	CaO	$\text{Ca}(\text{OH})_2$	$\text{CaOCl}_2 \cdot \text{H}_2\text{O}$
(2)	CaO	$\text{Ca}(\text{OCl})_2$	$\text{Ca}(\text{OH})_2$
(3)	$\text{Ca}(\text{OCl})_2$	$\text{Ca}(\text{OH})_2$	CaO
(4)	$\text{Ca}(\text{OH})_2$	CaO	$\text{Ca}(\text{OCl})_2$

133. Identify the correct set of 13th group elements which do not form amphoteric oxides

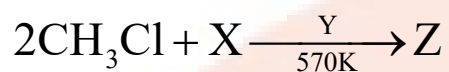
(1) B, In, Tl

(2) B, Al, Ga

(3) Al, Ga, Tl

(4) Al, Tl, In

134. Identify X, Y and Z in the following reaction



X

Y

Z

(1) C Ni $(\text{CH}_3)_2\text{Si}(\text{OH})_2$

(2) Si Zn $(\text{CH}_3)_2\text{SiCl}_2$

(3) Si Cu $(\text{CH}_3)_2\text{SiCl}_2$

(4) H_2O Si $(\text{CH}_3)_2\text{Si}(\text{OH})_2$

135. Which of the following is **not** a greenhouse gas?

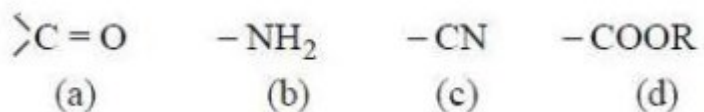
(1) CO_2

(2) O_3

(3) CH_4

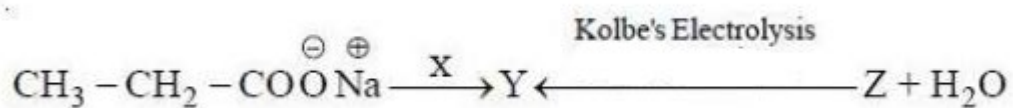
(4) N_2

136. The order of priority of the following functional groups in IUPAC method of naming organic compounds is



- (1) b, a, d, c
- (2) c, d, b, a
- (3) d, c, a, b
- (4) a, c, d, b

137. What are X, Y and Z in the following reactions?



X

Y

Z

- | | | | |
|-----|---|--|--|
| (1) | $\text{NaOH} + \text{CaO}/\Delta$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ | $\text{CH}_3\text{CH}_2\overset{\ominus}{\text{C}}\overset{\oplus}{\text{O}}\text{Na}$ |
| (2) | Mo_2O_3 | C_2H_6 | $\text{CH}_3\text{CH}_2\overset{\ominus}{\text{C}}\overset{\oplus}{\text{O}}\text{Na}$ |
| (3) | $\text{NaOH} + \text{CaO}/\Delta$ | C_2H_6 | $\text{CH}_3\overset{\ominus}{\text{C}}\overset{\oplus}{\text{O}}\text{Na}$ |
| (4) | $(\text{CH}_3\text{COO})_2\text{Mn}/\Delta$ | C_3H_8 | $\text{CH}_3\text{CH}_2\overset{\ominus}{\text{C}}\overset{\oplus}{\text{O}}\text{Na}$ |

138. Which one of the following compounds will **not** show geometrical isomerism?

- (1) prop-2-enoic acid

- (2) 2-butene
- (3) 2-methyl-2-butenic acid
- (4) 3-methyl-2-pentenoic acid

139. A compound is formed by X and Y elements. Atoms of Y (anions) form hcp lattice. Atoms of X (cations) are in some octahedral holes. The formula of the compound is XY_3 . What is the fraction of octahedral holes unoccupied by X?

- (1) $\frac{1}{2}$
- (2) $\frac{2}{3}$
- (3) $\frac{3}{4}$
- (4) $\frac{1}{5}$

140. At T(K), the vapour pressure of pure benzene is 0.85 bar. A non-volatile, non-electrolyte substance weighing 0.5 g when added to 39 g of benzene, the vapour pressure of the solution is 0.845 bar. The molar mass (in g mol^{-1}) of the substance is

- (1) 180

(2) 270

(3) 160

(4) 169

141. 0.1 m solution each of sodium sulphate, urea and sodium chloride are taken. The correct ratio of elevation of boiling point of these solutions is

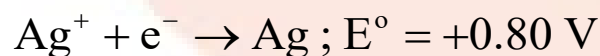
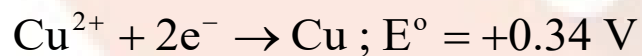
(1) 1:1:1

(2) 3:1:2

(3) 1:2:3

(4) 2:3:1

142. Using the standard electrode potentials given below identify the correct statements from the following



a) Copper can displace iron from FeSO_4 solution

b) Iron can displace copper from CuSO_4 solution

c) Silver can displace copper from CuSO_4 solution

d) Iron can displace silver from AgNO_3 solution

- (1) a, b
- (2) b, c
- (3) b, d
- (4) a, d

143. At T(K) if the rate constant for a zero order reaction is $2.5 \times 10^{-3} \text{ Ms}^{-1}$, the time required for the initial concentration of reactant, R to fall from 0.10 M to 0.075 M at the same temperature in seconds is

- (1) 25
- (2) 5
- (3) 10
- (4) 20

144. The temperature above which, formation of micelles takes place is called

- (1) Boyle's temperature
- (2) Kraft temperature
- (3) Critical temperature
- (4) Inversion temperature

145. The method used for producing semiconductor grade metals of high purity is

- (1) Poling
- (2) Electrolysis
- (3) Zone refining
- (4) Vapour phase refining

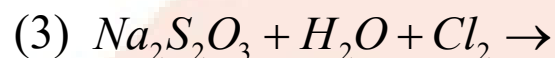
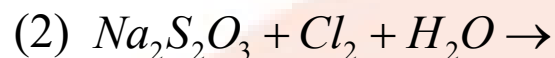
146. The element X on reaction with conc. HNO_3 forms two acidic oxides A and B of different shapes. Element Z on reaction with conc. H_2SO_4 forms two acidic oxides A and D of different shapes. What are X and Z?

- (1) C, C
- (2) S, Cu
- (3) C, S
- (4) C, Cu

147. Identify the correct statement from the following

- (1) O_3 , SO_2 molecules have different shapes
- (2) The molecular formula of pyrophosphoric acid is $\text{H}_2\text{S}_2\text{O}_8$.
- (3) In the presence of moisture, SO_2 acts as an oxidizing agent.
- (4) V_2O_5 acts as catalyst in contact process

148. Identify the reaction in which monobasic and dibasic acids are formed



149. Catalyst used in the manufacture of high density polythene is



150. Identify the correct statements from the following

a) Eu^{2+} and Yb^{2+} are reducing agents

b) The electronic configuration of Pr^{3+} is $[Xe]4f^3$

c) Aqueous solution of $LaCl_3$ is colourless

(1) a, b, c

(2) a, c

(3) a, b

(4) b, c

151. The polydispersity index of a polymer containing 10 molecules with molecular mass 1.0×10^4 and 10 molecules with molecular mass 1.0×10^5 is approximately

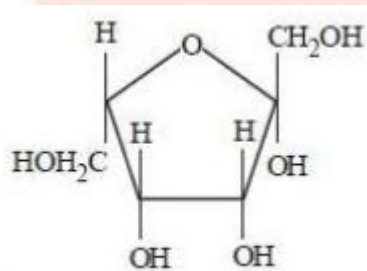
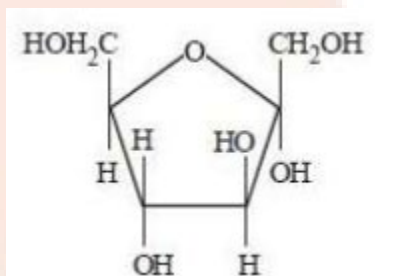
(1) 1.67

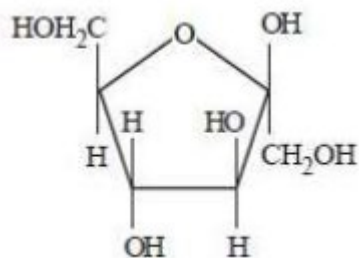
(2) 0.59

(3) 1.55

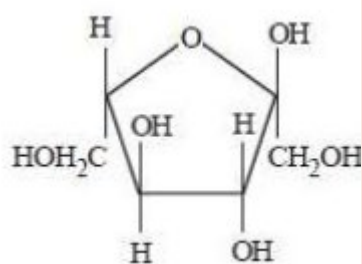
(4) 0.83

152. β -D-(-)-Fructofuranose is





(3)



(4)

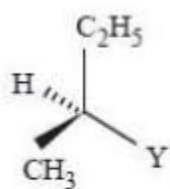
153. Assertion (A) : Shape of the receptor changes after attachment of chemical messenger

Reason (R) : Receptor does not regain original shape after removal of chemical messenger

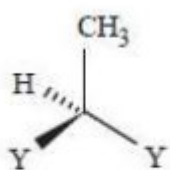
The correct answer is

- (1) (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

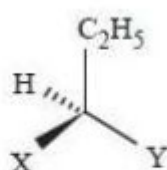
154. Identify chiral molecules from the following



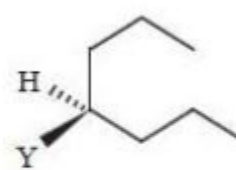
a



b



c



d

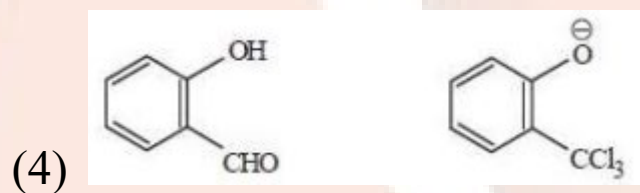
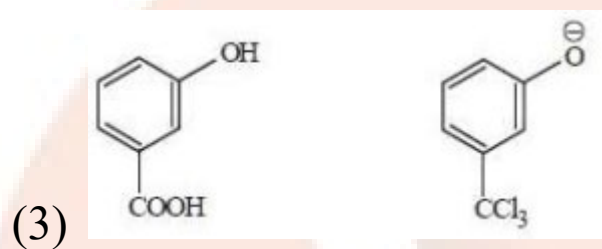
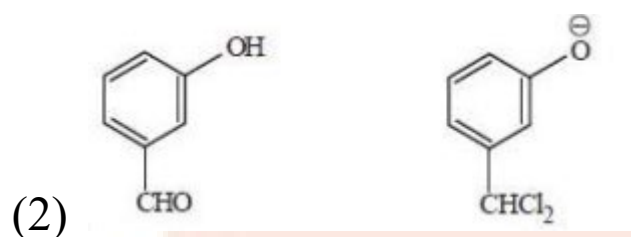
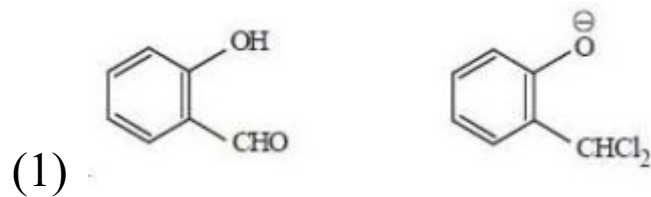
- (1) a, c
- (2) b, c, d
- (3) b, c
- (4) b, d

155. Isopropyl benzene on aerial oxidation followed by acid hydrolysis of the resulting compound yields

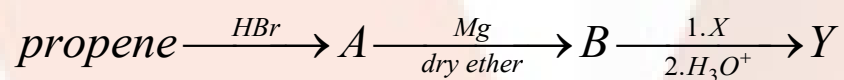
- (1) $(\text{CH}_3)_2\text{CO}$, $\text{C}_6\text{H}_5 - \text{C}_6\text{H}_5$
- (2) $(\text{CH}_3)_2\text{CO}$, $\text{C}_6\text{H}_5\text{CH}_3$
- (3) $(\text{CH}_3)_2\text{CO}$, $\text{C}_6\text{H}_5\text{OH}$
- (4) $(\text{CH}_3)_2\text{CHOH}$, $\text{C}_6\text{H}_5\text{CH}_3$

156. Reimer-Tiemann reaction involves the formation of X from phenol through intermediate Y. What are X and Y?

XY

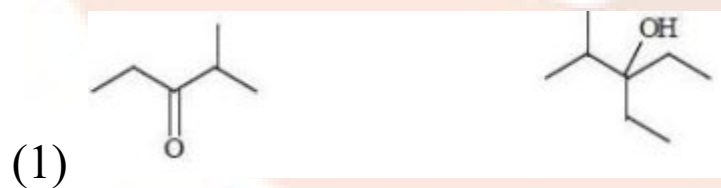


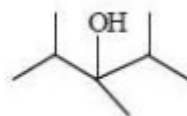
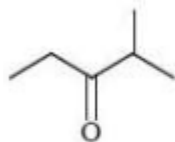
157. What are X and Y in the following reaction sequence?



X

Y





(4)

158. The hydrogen atom bonded to the carbon designated by which

number in $\overset{4}{\text{C}}\text{H}_3 - \overset{3}{\text{C}}\text{H}_2 - \overset{2}{\text{C}}\text{H}_2 - \overset{1}{\text{C}}\text{HO}$ is most acidic?

(1) C-4

(2) C-2

(3) C-3

(4) C-1

159. The order of strengths of the following carboxylic acids is

a) $\text{CH}_3 - \text{CH}_2 - \text{COOH}$

b) $\text{CH}_3 - \text{COOH}$

c) $\text{C}_6\text{H}_5 - \text{COOH}$

d) $\text{C}_6\text{H}_5 - \text{CH}_2\text{COOH}$

(1) $c > d > b > a$

(2) $d > b > c > a$

(3) $c > b > d > a$

(4) $a > d > b > c$

160. What are A, B and C in the following reaction?



- | <u>A</u> | <u>B</u> | <u>C</u> |
|-----------------------------|----------------|-----------------|
| (1) $\text{ArN}=\text{NAr}$ | N_2 | NaBF_4 |
| (2) ArNO_2 | N_2 | NaBF_4 |
| (3) ArNO | N_2 | NaBF_4 |
| (4) ArF | NaN_3 | BF_3 |

AP EAMCET-2018-Engineering-22 Apr-Shift-2
Answer Key

1	1	33	2	65	1	97	3	129	2
2	3	34	3	66	4	98	4	130	1
3	3	35	3	67	2	99	4	131	3
4	1	36	2	68	2	100	2	132	1
5	1	37	1	69	1	101	4	133	1
6	4	38	2	70	3	102	2	134	3
7	2	39	2	71	2	103	3	135	4
8	4	40	4	72	4	104	1	136	3
9	2	41	1	73	4	105	3	137	3
10	2	42	4	74	2	106	1	138	1
11	2	43	2	75	3	107	4	139	2
12	2	44	4	76	2	108	2	140	4
13	1	45	3	77	3	109	2	141	2
14	2	46	3	78	2	110	3	142	3
15	1	47	3	79	3	111	1	143	3
16	3	48	4	80	4	112	4	144	2
17	1	49	3	81	2	113	1	145	3
18	1	50	2	82	4	114	4	146	1

19	1	51	1	83	1	115	4	147	4
20	4	52	2	84	4	116	1	148	4
21	2	53	2	85	2	117	3	149	3
22	2	54	1	86	2	118	1	150	2
23	3	55	4	87	4	119	2	151	1
24	2	56	2	88	1	120	2	152	3
25	1	57	4	89	1	121	1	153	3
26	1	58	2	90	3	122	1	154	1
27	2	59	1	91	2	123	2	155	3
28	3	60	3	92	*	124	2	156	1
29	1	61	2	93	1	125	3	157	3
30	1	62	3	94	1	126	2	158	2
31	3	63	2	95	2	127	3	159	1
32	4	64	3	96	2	128	3	160	2

AP EAMCET-2018-Engineering-22 Apr-Shift-2
Solutions – Mathematics

1. Consider $f(x) = y$, so

$$\sin\left(g(x) - \frac{\pi}{3}\right) = \frac{y}{2}\sqrt{4-y^2}$$

and

$$\frac{y}{2}\sqrt{4-y^2} = t$$

$$4y^2 - y^4 = 4t^2$$

$$(y^2 - 2)^2 = -4t^2 + 4$$

$$t^2 = 1 - \frac{1}{4}(y^2 - 2)^2$$

As,

$$f(x) \in [-1, 1]$$

$$y^2 \in [0, 1]$$

$$t^2 \in \left[0, \frac{3}{4}\right]$$

$$t \in \left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$$

Thus,

$$g(x) - \frac{\pi}{3} \in \left[-\frac{\pi}{3}, \frac{\pi}{3} \right]$$

$$g(x) \in \left[0, \frac{2\pi}{3} \right]$$

2. The given function is define if,

$$\frac{4 - x^2}{[x] + 2} \geq 0$$

$$\Rightarrow \frac{x^2 - 4}{[x] + 2} \leq 0$$

So, either $x^2 - 4 \leq 0$ and

$$[x] + 2 > 0 \text{ or } x^2 - 4 \geq 0$$

For that

$$x \in [-2, 2]$$

$$x \in [-1, \infty)$$

$$\text{So, } x \in [-1, 2]$$

And $[x] + 2 < 0$, for which

$$x \in (-\infty, -2) \cup [-2, -\infty] \text{ and } x \in (-\infty, -2)$$

$$\text{So, } x \in (-\infty, -2)$$

From the intervals,

$$x \in (-\infty, -2) \cup [-1, 2]$$

3. On limiting case $a_n < a_\infty$

$$a_\infty = \sqrt{7 - a_\infty}$$

$$a_\infty^2 - a_\infty = 7$$

$$a_\infty = \frac{1 + \sqrt{29}}{2} \quad (\text{As, } a_\infty > 0)$$

So,

$$a_n < \frac{1 + \sqrt{29}}{2}, \forall n \geq 1$$

$$a_n < 4, \forall n \geq 1$$

4. Given: $A^2 + A + 2I = 0$

$$|A(A + I)| = |-2I|$$

$$|A||A + I| = (-2)^3$$

$$|A||A + I| = -8$$

$$|A| \neq 0, \text{ and } |A + I| \neq 0$$

Here, $|A| \neq 0$ so, A is not skew-symmetric matrix.

5. For square matrix of order 3,

$$\begin{aligned} |\text{Adj } A| &= |A|^{3-1} \\ &= |A|^2 \end{aligned}$$

If $|A| = 0$ then $|\text{Adj } A| = 0$ and

$$|A \cdot A^{-1}| = |I|, \text{ if } |A| \neq 0$$

So,

$$|A \cdot A^{-1}| = |I|$$

$$|A| |A^{-1}| = 1$$

$$|A^{-1}| = |A|^{-1}$$

Thus, statements I and II are correct.

6. Given equation,

$$x - 2y + 3z = 5$$

$$2x - 2y + z = 0$$

$$-z + 2y - 3z = 6$$

The determinant is,

$$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ 2 & -2 & 1 \\ -1 & 2 & -3 \end{vmatrix}$$

$$= 1(6 - 2) + 2(-6 + 1) + 3(4 - 2)$$

$$= 0$$

$$\Delta_1 = \begin{vmatrix} 5 & -2 & 3 \\ 0 & -2 & 1 \\ 6 & 2 & -3 \end{vmatrix}$$

$$= 5(6 - 2) + 2(0 - 6) + 3(0 - 12)$$

$$\neq 0$$

Thus, the system has no solution.

7. Given,

$$= \sin \frac{\pi}{5} + i \left(1 - \cos \frac{\pi}{5} \right) = 2 \sin \frac{\pi}{10} \cos \frac{\pi}{10} + i \left(2 \sin^2 \frac{\pi}{10} \right)$$

$$= 2 \sin \frac{\pi}{10} \left[\cos \frac{\pi}{10} + i \sin \frac{\pi}{10} \right]$$

So, the amplitude of the complex number is $\frac{\pi}{10}$.

8. Given,

$$\frac{z+1}{z+i} = \frac{x+iy+1}{x+iy+i}$$

$$= \frac{(x+1)+iy}{x+i(y+1)} \times \frac{x-i(y+1)}{x-i(y+1)}$$

For $\frac{z+1}{z+i}$ to be purely real number, $z \neq -i$

So,

$$I_m\left(\frac{z+1}{z+i}\right) = 0$$

$$z = -i$$

$$xy - (x+1)(y+1) = 0$$

$$xy - xy - x - y - 1 = 0$$

$$x + y + 1 = 0$$

$$(x, y) \neq (0, -1)$$

9. Given,

$$\left[\frac{51+73\omega+87\omega^2}{73+87\omega+51\omega^2} + \frac{51+73\omega+87\omega^2}{87+51\omega+73\omega^2} \right]^{15}$$

Solve further,

$$= \left[\frac{51\omega^2 + 73\omega^3 + 87\omega^4}{73+87\omega+51\omega^2} \times \frac{1}{\omega^2} + \frac{51\omega + 73\omega^2 + 87\omega^3}{87+51\omega+73\omega^2} \times \frac{1}{\omega} \right]^{15}$$

$$= (\omega + \omega^2)^{15}$$

$$= (-1)^{15}$$

$$= -1$$

10. For the given equation,

$$iz^3 + 4z^2 - z + 4i = 0$$

$$z^2(iz + 4) + i(iz + 4) = 0$$

$$z = 4i, z^2 = i$$

$$z = 4i$$

So,

$$z = \pm \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}i \right)$$

So, minimum magnitude of the complex root,

$$\frac{1-i}{\sqrt{2}}, \text{ or } \frac{-1+i}{\sqrt{2}}$$

11. The roots of the equation is α and β .

So,

$$\alpha + \beta = 4$$

$$\alpha\beta = 5$$

Now,

$$\begin{aligned} (\alpha^2 + \beta) + (\alpha + \beta^2) &= (\alpha + \beta)^2 - 2\alpha\beta + (\alpha + \beta) \\ &= 16 - 10 + 4 \\ &= 10 \end{aligned}$$

And

$$\begin{aligned}
 (\alpha^2 + \beta)(\alpha + \beta^2) &= \alpha^3 + \alpha^2\beta^2 + \beta\alpha + \beta^3 \\
 &= \alpha^3 + \beta^3 + \alpha\beta(\alpha\beta + 1) \\
 &= (\alpha + \beta)\left[(\alpha + \beta)^2 - 3\alpha\beta\right] + \alpha\beta(\alpha\beta + 1) \\
 &= 4[16 - 15] + 5(6) \\
 &= 34
 \end{aligned}$$

So, the quadratic equation whose roots are $(\alpha^2 + \beta)$ and

$(\alpha + \beta^2)$ is

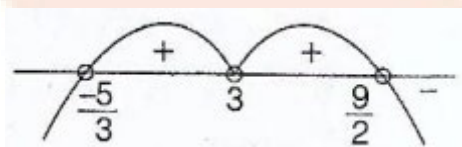
$$x^2 - (\alpha^2 + \beta + \alpha + \beta^2)x + (\alpha^2 + \beta)(\alpha + \beta^2) = 0$$

$$x^2 - 10x + 34 = 0$$

12. $f(x) \cdot g(x) = ab$

$$\left(x + \frac{5}{3}\right)(3 - x)(x - 3)\left(x - \frac{9}{2}\right)$$

From wave curve method,



So, $f(x) \cdot g(x)$ is found positive in range $[0, 3) \cup \left(3, \frac{9}{2}\right)$ and

found negative in range $\left[\frac{9}{2}, 5\right]$.

13. As the given equation has no real roots and $4a + 2b + c > 0$,

Then, $4a + 2b + c > 0$

So,

$$a + b + c > 0$$

So,

$$c + a > -b$$

$$c + b > -a$$

Thus,

$$(c + a)(c + b) > ab.$$

14. The polynomial equation has two roots $\sqrt{3} + \sqrt{27}$ and

$\sqrt{2} + 5i$. So, other roots are,

$$\sqrt{3} - \sqrt{27} \text{ and } \sqrt{2} - 5i$$

$$-\sqrt{3} + \sqrt{27} \text{ and } -\sqrt{2} - 5i$$

Thus, the number of roots of the equation is six and degree of the equation having six distinct roots is six.

15. Total numbers that starts with digit 6, 5 and 4 is 24, 24 and 24 respectively.

So, total numbers are $24 + 24 + 24 = 72$

Now, total number that starts with 36, 35, and 34 are 6, 6, and 6 respectively.

The 90th number is 34256.

Therefore, 89th number is 34265.

16. If one vertex at each line then number of triangles are,

$$n_1 = {}^5C_1 \times {}^7C_1 \times {}^9C_1$$

$$n_1 = 315$$

If two vertex at line L_1 and remaining ether at L_2 and L_3 , then number of triangles are,

$$n_2 = {}^5C_2 \times {}^{16}C_1$$

$$n_2 = 160$$

If two vertex at line L_2 and remaining ether at L_1 and L_3 , then number of triangles are,

$$n_3 = {}^7C_2 \times {}^{14}C_1$$

$$n_3 = 294$$

If two vertex at line L_3 and remaining ether at L_1 and L_2 , then number of triangles are,

$$n_4 = {}^9C_2 \times {}^{12}C_1$$

$$n_4 = 432$$

17. The expression of the general term is,

$$\begin{aligned} T_{r+1} &= {}^6C_r (ax^3)^{6-r} \left(\frac{c}{x}\right)^r \\ &= {}^6C_r a^{6-r} c^r x^{18-4r} \end{aligned}$$

According to the equation,

$$18 - 4r = 2$$

$$r = 4$$

So,

$${}^6C_4 a^{6-4} c^4 = 60$$

$$15a^2 c^4 = 60$$

$$ac^2 = \pm 2$$

$$a > 0$$

$$ac^2 = 2$$

18. Let,

$$(1+y)^n = 1 + ny + \frac{n(n-1)}{2!} y^2 + \frac{n(n-1)(n-2)}{3!} y^3 + \dots$$

Compare the up to third term of given series with above series,

$$\frac{n(n-1)}{2!} y^2 = \frac{3}{4.8} \quad \text{and} \quad \frac{n(n-1)(n-2)}{3!} y^3 = \frac{3 \cdot 5}{4 \cdot 8 \cdot 12}$$

And

$$\frac{n(n-1)(n-2)(n-3)}{4!}y^4 = \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12 \cdot 16}$$

By solving the equations,

$$\begin{aligned} x &= (1+y)^n - 1 - ny \\ &= \left(1 - \frac{1}{2}\right)^{-1/2} - 1 - \left(-\frac{1}{2}\right)\left(\frac{-1}{2}\right) \\ &= \sqrt{2} - \frac{5}{4} \end{aligned}$$

Now,

$$\begin{aligned} 2x^2 + 5x &= 2\left[2 + \frac{25}{16} - \frac{5}{\sqrt{2}}\right] + 5\sqrt{2} - \frac{25}{4} \\ &= 4 + \frac{25}{8} - 5\sqrt{2} + 5\sqrt{2} - \frac{25}{4} \\ &= \frac{7}{8} \end{aligned}$$

19. If,

$$\frac{3x^2 + 1}{(x^2 + 1)(x^2 + 2)^2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 2} + \frac{Ex + F}{(x^2 + 2)^2}$$
$$= \frac{\left[Ax + B(x^2 + 2)^2 + (Cx + D)(x^2 + 1)(x^2 + 2) + (Ex + F)(x^2 + 1) \right]}{(x^2 + 1)(x^2 + 2)^2}$$

$$3x^2 + 1 = \frac{\left[(Ax + B)(x^2 + 2)^2 + (Cx + D)(x^2 + 1)(x^2 + 2) + (Ex + F)(x^2 + 1) \right]}{(x^2 + 1)(x^2 + 2)^2}$$

Compare coefficient of different terms,

$$A = 0$$

$$C = 0$$

$$E = 0$$

So,

$$A + C + E = 0$$

20. Given: $\tan\left(\frac{\pi}{4} + \frac{y}{2}\right) = \tan^3\left(\frac{\pi}{4} + \frac{x}{2}\right),$

$$\frac{1 + \tan \frac{y}{2}}{1 - \tan \frac{y}{2}} = \left(\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right)^3$$

$$\frac{\cos \frac{y}{2} + \sin \frac{y}{2}}{\cos \frac{y}{2} - \sin \frac{y}{2}} = \left(\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right)^3$$

Take square both sides,

$$\frac{1 + \sin y}{1 - \sin y} = \left(\frac{1 + \sin x}{1 - \sin x} \right)^3$$

Apply component and dividend rule,

$$\frac{2}{2 \sin y} = \frac{(1 + \sin x)^3 + (1 - \sin x)^3}{(1 + \sin x)^3 - (1 - \sin x)^3}$$

$$\frac{3 \sin x + \sin^3 x}{1 + 3 \sin^2 x} = \sin y$$

21. Solve the given expression,

$$\begin{aligned}
 &= \left[(\cos^2 252^\circ - \sin^2 126^\circ)(\cos 252^\circ + \sin 126^\circ) \right] \\
 &= \left[(\sin^2 126^\circ + \sin^2 186^\circ + \sin^2 66^\circ) \right] \\
 &= \left[\left[\frac{1 + \cos 504^\circ}{2} - \frac{1 - \cos 252^\circ}{2} \right] \right] \\
 &= \left[\left[\frac{1 - \cos 252^\circ + 1 - \cos 372^\circ + 1 - \cos 132^\circ}{2} \right] \right] \\
 &= \frac{1}{4} \left[(\cos 504^\circ + \cos 252^\circ) \right] \\
 &= \frac{1}{4} \left[(3 - \cos 252^\circ - 2 \cos 252^\circ \cos 120^\circ) \right] \\
 &= \frac{3}{2} \cos 18^\circ (-\sin 36^\circ) \\
 &= -\frac{3\sqrt{5}}{8}
 \end{aligned}$$

22. Solve the given expression,

$$\begin{aligned}
 &\cos \alpha + \cos \beta - \cos \gamma - \cos(\alpha + \beta + \gamma) \\
 &= 2 \sin \frac{2\alpha + \beta + \gamma}{2} \sin \frac{\beta + \gamma}{2} + 2 \sin \frac{\beta + \gamma}{2} \sin \left(\frac{\gamma - \beta}{+2} \right) \\
 &= 4 \sin \frac{\beta + \gamma}{2} \sin \frac{\alpha + \gamma}{2} \cos \frac{\alpha + \beta}{2} \\
 &= 4 \cos \frac{\alpha + \beta}{2} \sin \frac{\beta + \gamma}{2} \sin \frac{\gamma + \alpha}{2}
 \end{aligned}$$

23. Solving the given expression,

$$\sqrt{3 - 5\sin x + \sin^2 x} = -\cos x$$

$$3 - 5\sin x + \sin^2 x = \cos^2 x$$

$$2\sin x(\sin x - 2) - 1(\sin x - 2) = 0$$

$$\sin x = \frac{1}{2} \text{ or } 2$$

$$\sin x \in [-1, 1]$$

$$\text{So, } \sin x = \frac{1}{2} \text{ and } \cos < 0$$

It means,

$$\sin x = \frac{1}{2}, \cos x = -\frac{\sqrt{3}}{2}$$

$$\text{Thus, } x = (2n+1)\pi - \frac{\pi}{6}, n \in \mathbb{Z}$$

24. From given statements,

$$\sin^{-1}(y^2 - 4y + 6) + \cos^{-1}(y^2 - 4y + 6) = \frac{\pi}{2} \quad \dots\dots (I)$$

$$\sec^{-1}(y^2 - 4y + 6) + \operatorname{cosec}^{-1}(y^2 - 4y + 6) = \frac{\pi}{2} \quad \dots\dots (II)$$

$$\text{As } y^2 - 4y + 6 = (y - 2)^2 + 2 \geq 2, \forall y \in \mathbb{R}.$$

So, statement (I) is false and statement (II) is true.

25. Given,

$$\sec \theta \cosh y = \operatorname{cosec} x$$

$$\cos x = \sin x \cosh y \quad \dots\dots (I)$$

And

$$\operatorname{cosec} \theta \sinh y = \sec x$$

$$\sin \theta = \cos x \sinh y \quad \dots\dots (II)$$

Square and add both equations,

$$1 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$

$$\sin^2 x (1 + \sinh^2 y) + \cos^2 x \sinh^2 y = 1$$

$$\sinh^2 y + \sin^2 x = 1$$

$$\sinh^2 y = \cos^2 x$$

26. From given statement (I),

$$\sin C = \sqrt{1 - \frac{121}{625}}$$

$$= \sqrt{\frac{504}{25}}$$

$$= \frac{6\sqrt{14}}{25}$$

So,

$$\frac{c}{\sin C} = 2R$$

$$\begin{aligned} R &= \frac{6}{2\left(\frac{6\sqrt{14}}{25}\right)} \\ &= \frac{25}{2\sqrt{14}} \end{aligned}$$

So, statement (I) is true.

From statement (II),

$$\begin{aligned} \cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{9 + 16 - 36}{2(3)(6)} \\ &= \frac{-11}{24} \end{aligned}$$

$$\cos C < 0$$

So, the triangle is obtuse angled triangle.

Thus, the statement (II) is false.

27. As,

$$\begin{aligned}\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} &= \left[\frac{s(s-a)}{\Delta} + \frac{s(s-b)}{\Delta} + \frac{s(s-c)}{\Delta} \right] \\ &= \frac{S}{\Delta} [3S - (a+b+c)] \\ &= \frac{(a+b+c)^2}{4\Delta}\end{aligned}$$

Now,

$$\begin{aligned}\cot A + \cot B + \cot C &= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} \\ &= \frac{b^2 + c^2 - a^2}{2bc \sin A} + \frac{c^2 + a^2 - b^2}{2ac \sin B} + \frac{a^2 + b^2 - c^2}{2ab \sin C} \\ &= \frac{a^2 + b^2 + c^2}{4\Delta}\end{aligned}$$

So,

$$\begin{aligned}\frac{\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}}{\cot A + \cot B + \cot C} &= \frac{(a+b+c)^2}{a^2 + b^2 + c^2} \\ &= \frac{(3+4+6)^2}{9+16+36} \\ &= \frac{169}{61}\end{aligned}$$

28. As $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in H.P., then a, b , and c are in A.P.

So, $s - a, s - b, s - c$ are in A.P.

Hence, $\frac{s - a}{\Delta}, \frac{s - b}{\Delta}, \frac{s - c}{\Delta}$ are in A.P.

So, r_1, r_2, r_3 are in H.P.

29. Consider the point $P(p, 2^{p+2})$ and $Q(q, 2^{q+2})$.

Now,

$$OP = Pi + 2^{p+2} j$$

$$OQ = qi + 2^{q+2} j$$

From question,

$$OP \cdot i = -1$$

$$p = -1$$

$$\Rightarrow OP = -\hat{i} + 2\hat{j}$$

And,

$$OQ \cdot i = 2$$

$$q = 2$$

$$\Rightarrow OQ = 2\hat{i} + 16\hat{j}$$

So,

$$\text{OQ} - 4\text{OP} = 6\hat{i} + 8\hat{j}$$

$$|\text{OQ} - 4\text{OP}| = \sqrt{36 + 64} \\ = 10$$

30. Equation of line is,

$$\frac{x-3}{\frac{2}{3}} = \frac{y-1}{\frac{-1}{3}} = \frac{z+1}{\frac{2}{3}} = \pm 3$$

Hence, the coordinates of P and Q are (5, 0, 1) and (1, 2, -3).

Now, equation of the plane OPQ is,

$$r = s(\text{OQ}) + y(\text{OP})$$

As r is coplanar with vectors OP and OQ.

$$r = (s + 5t)\hat{i} + 2s\hat{j} + (t - 3s)\hat{k}$$

31. For vector **m** which coplanar with $(2\hat{i} + \hat{j})$ and $(2\hat{i} + \hat{j})$

So,

$$\mathbf{m} = x(2\hat{i} + \hat{j}) + y(\hat{j} - \hat{k}) \\ = 2x\hat{i} + (x + y)\hat{j} - y\hat{k}$$

m is orthogonal to the vector $\hat{i} - \hat{j} + \hat{k}$, so

$$2x - (x - y) - y = 0$$

$$x = 2y$$

So,

$$|\mathbf{m}| = 1$$

$$4x^2 + (x + y)^2 + y^2 = 1$$

$$26y^2 = 1$$

$$y = \pm \frac{1}{\sqrt{26}} \Rightarrow x = \pm \frac{2}{\sqrt{26}}$$

So, the vector $\mathbf{m}$,

$$\mathbf{m} = \pm \left(\frac{4}{\sqrt{26}} \hat{\mathbf{i}} + \frac{3}{\sqrt{26}} \hat{\mathbf{j}} - \frac{1}{\sqrt{26}} \hat{\mathbf{k}} \right)$$

The length is perpendicular from origin to plane, $\mathbf{r} \cdot \mathbf{m} = \mathbf{a} \cdot \mathbf{m}$.

$$\begin{aligned} |\mathbf{a} \cdot \mathbf{m}| &= \left| (\hat{\mathbf{i}} - \hat{\mathbf{k}}) \cdot \left[\frac{4}{\sqrt{26}} \hat{\mathbf{i}} + \frac{3}{\sqrt{26}} \hat{\mathbf{j}} - \frac{1}{\sqrt{26}} \hat{\mathbf{k}} \right] \right| \\ &= \frac{4}{\sqrt{26}} + \frac{1}{\sqrt{26}} \\ &= \frac{5}{\sqrt{26}} \end{aligned}$$

So,

32. Since,

$$\begin{aligned}\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{a} \cdot \mathbf{c}) \cdot \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c} \\ &= \frac{\mathbf{b} + \mathbf{c}}{\sqrt{2}}\end{aligned}$$

So,

$$\mathbf{a} \cdot \mathbf{b} = \frac{-1}{\sqrt{2}}$$

$$\cos \theta = -\frac{1}{\sqrt{2}}$$

$$\theta = \frac{3\pi}{4}$$

33. $\mathbf{c} = (\mathbf{a} \times \mathbf{c}) + \mathbf{b}$

$$|\mathbf{c}|^2 = |\mathbf{a} + \mathbf{c}|^2 + |\mathbf{b}|^2 + 2(\mathbf{a} \times \mathbf{c}) \cdot \mathbf{b}$$

$$[\mathbf{a} \mathbf{b} \mathbf{c}] = \frac{1}{2} [1 - |\mathbf{c}|^2 \cos^2 \theta]$$

Now,

$$\mathbf{c} \cdot (\mathbf{a} \cdot \mathbf{c}) = (\mathbf{a} \times \mathbf{c})^2 + \mathbf{b} \cdot (\mathbf{a} \times \mathbf{c})$$

$$[\mathbf{a} \mathbf{b} \mathbf{c}] = |\mathbf{c}|^2 \sin^2 \theta$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta + 2 \sin^2 \theta}$$

$$= \frac{1}{\operatorname{cosec}^2 \theta + 1}$$

For maximum value of $[\mathbf{a} \mathbf{b} \mathbf{c}]$, $\operatorname{cosec}^2 \theta = 1$. So,

$$[\mathbf{a} \mathbf{b} \mathbf{c}]_{\max} = \frac{1}{1+1}$$

$$= \frac{1}{2}$$

34. Solve the expression,

$$\begin{aligned}
 & (\alpha \times (\beta + \gamma)) \times (\beta \times \gamma) \cdot (\beta - \gamma) \\
 &= (\alpha \times (\beta + \gamma)) \cdot [(\beta \times \gamma) \times (\beta - \gamma)] \\
 &= (\alpha \times (\beta + \gamma)) \cdot [(|\beta|^2 - \beta \cdot \gamma) \gamma - (\gamma \cdot \beta - |\gamma|^2) \beta] \\
 &= (1 - \beta \cdot \gamma) [(\alpha \times \beta) \cdot \gamma + (\alpha \times \gamma) \cdot \beta] \\
 &= (1 - \beta \cdot \gamma) [(\alpha \times \beta) \cdot \gamma - (\alpha \times \beta) \cdot \gamma] \\
 &= 0
 \end{aligned}$$

35. Solve as follows,

$$\frac{\sum_{i=1}^9 x_i}{9} = 13$$

$$\sum_{i=1}^9 x_i = 117$$

And

$$\sigma^2 = 25$$

$$\frac{\sum_{i=1}^9 x_i^2}{9} - (13)^2 = 25$$

$$\sum_{i=1}^9 x_i^2 = 9[25 + 169]$$

Including 10th item of value 3,

$$\begin{aligned}\bar{y} &= \frac{\sum_{i=1}^9 x_i + 3}{10} \\ &= \frac{117 + 3}{10} \\ &= 12\end{aligned}$$

And new variance is calculated as,

$$\begin{aligned}&= \frac{\left(\sum_{i=1}^9 x_i^2 + 9 \right)}{10} - (12)^2 \\ &= \frac{(169 + 25 + 1)9}{10} - 144 \\ &= \frac{1755 - 1440}{10} \\ &= 31.5\end{aligned}$$

36. From given value prepare the table,

Marks	Mid interval value (x)	f	f(x)	x ²	f(x ²)
1-3	2	40	80	4	160
3-5	4	30	120	16	480
5-7	6	20	120	36	720
7-9	8	10	80	64	640

So, the variance is,

$$V = \frac{\sum f \cdot x^2}{\sum f} - \left(\frac{\sum f \cdot x}{\sum f} \right)^2$$

$$V = 20 - 16$$

$$V = 4$$

37. Steps of the solution is given as follows,

$$\begin{aligned}
 P(\bar{A} \cap \bar{B}) &= P(A \cup B) \\
 &= 1 - P(P \cup B) \\
 &= 1 - \{P(A) + P(B) - P(A \cap B)\} \\
 &= 1 - \{0.58 + 0.32 + -0.28\} \\
 &= 1 - 0.62 \\
 &= 0.38
 \end{aligned}$$

38. For given conditions,

$$A = \left[\begin{array}{l} (2,6), (3,5), (4,4), (5,3), (6,2), \\ (3,6), (4,5), (5,4), (6,3), (4,6), \\ (5,5), (6,4), (5,6), (6,5), (6,6) \end{array} \right]$$

And event $A \cap B = (2,6), (3,5), (5,3), (6,2), (3,6), (6,3)$

So,

$$\begin{aligned} P(B/A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{6}{15} \text{ or } \frac{2}{5} \end{aligned}$$

39. Let,

E = drawn balls are red.

C = 6 red balls in bag.

A = 4 red balls in bag.

B = 5 red balls in bag.

Then,

$$P(A) = P(B) = P(C)$$

And

$$P\left(\frac{E}{A}\right) = \frac{{}^4C_4}{{}^6C_4}$$
$$= \frac{1}{15}$$

$$P\left(\frac{E}{B}\right) = \frac{{}^5C_4}{{}^6C_4}$$
$$= \frac{5}{15}$$

$$P\left(\frac{E}{C}\right) = \frac{{}^6C_4}{{}^6C_4}$$
$$= \frac{15}{15}$$

Apply Baye's theorem.

$$P\left(\frac{B}{A}\right) = \frac{P(B) \cdot P\left(\frac{E}{B}\right)}{P(A)P\left(\frac{E}{A}\right) + P(B)P\left(\frac{E}{B}\right) + P(C)P\left(\frac{E}{C}\right)}$$
$$= \frac{\frac{5}{15}}{\frac{1}{15} + \frac{5}{15} + \frac{15}{15}}$$
$$= \frac{5}{21}$$

40. The variance is,

$$V = E(X^2) - (E(X))^2$$

As,

$$\text{Sum of } P(x = x_i) = 1$$

$$\frac{1}{8} + \frac{3}{8} + 3k + k = 1$$

$$k = \frac{1}{8}$$

Now,

$$\begin{aligned} V &= \left[\left[\left(0^2 \times \frac{1}{8} \right) + \left(1^2 \times \frac{3}{8} \right) + \left(2^2 \times \frac{3}{8} \right) + \left(3^2 \times \frac{1}{8} \right) \right] \right. \\ &\quad \left. - \left[\left(0 \times \frac{1}{8} \right) + \left(1 \times \frac{3}{8} \right) + \left(2 \times \frac{3}{8} \right) + \left(3 \times \frac{1}{8} \right) \right]^2 \right] \\ &= \left[\frac{3}{8} + \frac{12}{8} + \frac{9}{8} \right] - \left[\frac{3}{8} + \frac{6}{8} + \frac{3}{8} \right]^2 \\ &= 3 - \frac{9}{4} \\ &= \frac{3}{4} \end{aligned}$$

41. The mean of defective locks m is calculated as,

$$\begin{aligned} m &= 100 \times \frac{2}{100} \\ &= 2 \end{aligned}$$

The probability is,

$$= 1 - [P(r=0) + P(r=1) + P(r=2)]$$

$$= 1 - \left[e^{-2} \frac{2^0}{0!} + e^{-2} \frac{2^1}{1!} + e^{-2} \frac{2^2}{2!} \right]$$

$$= 1 - 5e^{-2}$$

42. Let,

$$(2\cos\theta - 3, 3\sin\theta - 4) = (x, y)$$

So,

$$\cos\theta = \frac{x+3}{2}, \sin\theta = \frac{y+4}{3}$$

Now,

$$\left(\frac{y+4}{3}\right)^2 + \left(\frac{x+3}{2}\right)^2 = 1$$

$$4(y^2 + 8y + 16) + 9(x^2 + 6x + 9) = 1$$

$$9x^2 + 4y^2 + 54x + 32y + 109 = 0$$

43. Take $(x' + 2)$ for x and $(y' + 2)$ for y to change the origin.

So, new equation can be written as,

$$\left[\begin{array}{l} 3(x'+2)^2 + 2(x'+2)(y'+3) + 3(y'+3)^2 \\ -18(x'+2) - 22(y'+3) + 50 \end{array} \right] = 0$$

Solve and obtain,

$$3x'^2 + 2x'y' + 3y'^2 - 1 = 0$$

By dropping the suffices the equation can be written as,

$$3x^2 + 2xy + 3y^2 - 1 = 0$$

For to turn the axes through an angle of $\frac{\pi}{3}$, in the counter

clockwise sense, we should write $\frac{x' - \sqrt{3}y'}{2}$ for x and $\frac{\sqrt{3}x' + y'}{2}$

for y .

So,

$$\left[\begin{array}{l} 3\left(\frac{x' - \sqrt{3}y'}{2}\right)^2 + 2\left(\frac{x' - \sqrt{3}y'}{2}\right)\left(\frac{\sqrt{3}x' + y'}{2}\right) \\ + 3\left(\frac{\sqrt{3}x' + y'}{2}\right)^2 - 1 \end{array} \right] = 0$$

By dropping the suffices and solving the equation,

$$\left[3(x^2 + 3y^2 - 2\sqrt{3}xy) + 2(\sqrt{3}x^2 - 2xy - \sqrt{3}y^2) \right. \\ \left. + 3(3x^2 + y^2 + 2\sqrt{3}xy) - 4 \right] = 0$$

$$(12 + 2\sqrt{3})x^2 - 4xy + (12 - 2\sqrt{3})y^2 - 4 = 0$$

$$(6 + \sqrt{3})x^2 - 2xy + (6 - \sqrt{3})y^2 - 2 = 0$$

44. The equation of line passes through the point $(1, 1)$ of slope m is,

$$y - 1 = m(x - 1)$$

So, point $A\left(\frac{m-1}{m}, 0\right)$ and $B(0, 1-m)$.

Now,

$$OA + OB = \left(1 - \frac{1}{m}\right) + (1 - m) \\ = 2 - \left(m + \frac{1}{m}\right)$$

As m is negative, so minimum value of

$$-\left(m + \frac{1}{m}\right) = 2$$

Thus, minimum value of $OA + OB$.

$$OA + OB = 2 + 2 \\ = 4$$

45. The point of intersection of line $L = 0$ and line segment PQ is,

$$R\left(\frac{(4 \times 1) + (1 \times 2)}{5}, \frac{(4 \times 1) + (1 \times (-1))}{5}\right) = R\left(\frac{6}{5}, \frac{3}{5}\right)$$

For point R on line $L = 0$ is,

$$\frac{18}{5} + \frac{12}{5} - k = 0$$

$$k = 6$$

Now, equation of line concurrent with the lines $L = 0$ and PQ ,

$$(3x + 4y - 6) + \lambda(2x + y - 3) = 0$$

The above line is parallel to $x = y$, so

$$-\frac{3 + 2\lambda}{4 + \lambda} = 1$$

$$\lambda = -\frac{7}{3}$$

So, the equation of the line is,

$$5x - 5y - 3 = 0$$

46. The centroid of a non-equilateral triangle divides the line orthocenter and circumcentre in 2:1, thus, the coordinates of circumcentre is (h, k) , then

$$\left(\frac{2h+5}{3}, \frac{2k+8}{3}\right) = \left(3, \frac{14}{3}\right)$$

$$(h, k) = (2, 3)$$

The image of orthocenter $(5, 8)$ with respect to side BC ,

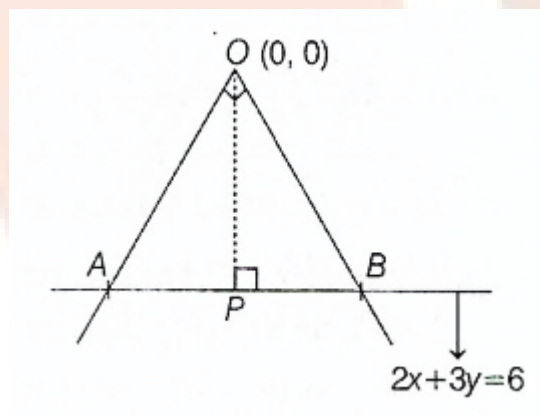
So, the radius of circumcircle of $\triangle ABC$ is

$$\sqrt{(8-2)^2 + (5-3)^2} = \sqrt{40} = 2\sqrt{10}$$

So, the diameter is $4\sqrt{10}$.

47. As, $\triangle AOB$ is a right angle triangle, the angle A and B is $\frac{\pi}{4}$ as

shown below.



So,

$$\begin{aligned} AB &= 2AP \\ &= 2OP \end{aligned}$$

Now,

$$OP = \frac{|6|}{\sqrt{4+9}}$$

$$= \frac{6}{\sqrt{13}}$$

Thus, the area of the triangle ΔAOB is,

$$= \frac{1}{2} \times AB \times OP$$

$$= OP^2$$

$$= \frac{36}{13}$$

48. The equation,

$$3x^2 + 7xy + 2y^2 + 2gx + 2fy + 2 = 0$$

Above equations shows the pair of intersecting lines, so

$$(3)(2)^2 + 7gf - 3f^2 - 2g^2 - 49 = 0$$

$$24 + 14gf - 6f^2 - 4g^2 - 49 = 0$$

$$4g^2 + 6f^2 - 14gf + 25 = 0$$

To get point of intersection of pair of straight line the equation should be differentiate with respect to x and y .

So,

$$6x + 7y + 2g = 0 \quad \dots\dots (I)$$

$$7x + 4y + 2f = 0 \quad \dots\dots (II)$$

Now, point of intersection of line (I) and (II) is,

$$\left(\frac{14f - 8g}{25}, \frac{14g - 12f}{25} \right)$$

From question,

$$\left(\frac{14f - 8g}{25} \right)^2 + \left(\frac{14g - 12f}{25} \right)^2 = \frac{2}{5}$$

$$\left[\begin{array}{l} 196f^2 + 64g^2 - 224fg + 196g^2 \\ + 144f^2 - 336gf \end{array} \right] = 250$$

$$26g^2 + 34f^2 - 56gf = 25$$

Now,

$$26g^2 + 34f^2 - 56gf = 25$$

$$16g^2 + 24f^2 - 56gf = -100$$

$$\begin{array}{ccccccc} - & - & + & + & & & \\ \hline \end{array}$$

$$10g^2 + 10f^2 = 125$$

$$g^2 + f^2 = \frac{25}{2}$$

49. Equation of pair of tangents from the point $(0, b)$ drawn to the circle $x^2 + y^2 = 16$ is,

$$(x^2 + y^2 - 16)(b^2 - 16) = (by + 16)^2$$

For point A and B , take $y = 0$,

So,

$$(x^2 - 16)(b^2 - 16) = 16^2$$

$$x = \pm \frac{4b}{\sqrt{b^2 - 16}}$$

Now,

For ΔPAB ,

$$\begin{aligned}\Delta &= \pm \frac{1}{2}b \left(\frac{8b}{\sqrt{b^2 - 16}} \right) \\ &= \frac{4b^2}{\sqrt{b^2 - 16}}\end{aligned}$$

For minimum area $\frac{d\Delta}{db} = 0$,

$$\sqrt{b^2 - 16}(8b) = 4b^2 \frac{b}{\sqrt{b^2 - 16}}$$

$$b^2 = 32$$

$$b = \pm 4\sqrt{2}$$

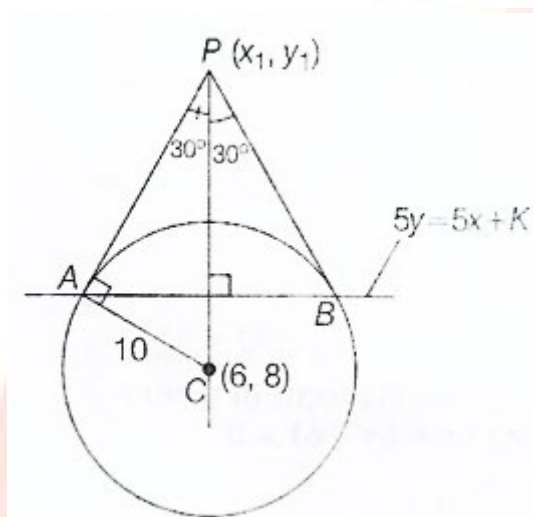
So, $P(0, \pm 4\sqrt{2})$, $A(4\sqrt{2}, 0)$ and $B(-4\sqrt{2}, 0)$

Thus, the equation of the circumference of ΔPAB is

$$x^2 + y^2 = 32$$

50. Let point $P(x_1, y_1)$, then

$$\frac{10}{\sqrt{x_1^2 + y_1^2 - 12x_1 - 16y_1}} = \frac{1}{\sqrt{3}}$$



From figure,

$$x_1^2 + y_1^2 - 12x_1 - 16y_1 = 300$$

$$(x_1 - 6)^2 + (y_1 - 8)^2 = 400 \quad \text{..... (I)}$$

As, line AB is a chord of contact of the circle with respect to point $P(x_1, y_1)$, so

$$xx_1 + yy_1 - 6(x + x_1) - 8(y + y_1) = 0$$

$$x(x_1 - 6) + y(y_1 - 8) - (6x_1 + 8y_1) = 0$$

$$\frac{x_1 - 6}{-5} = \frac{y_1 - 8}{5} = \frac{6x_1 + 8y_1}{K} = \lambda$$

So,

$$x_1 = 6 - 5\lambda, \text{ and } y_1 = 8 + 5\lambda \quad \text{..... (II)}$$

Form equation (I) and (II), we get

$$25\lambda^2 + 25\lambda^2 = 400$$

$$\lambda = \pm 2\sqrt{2}$$

$$\text{Also, } \lambda = \frac{6x_1 + 8y_1}{k}$$

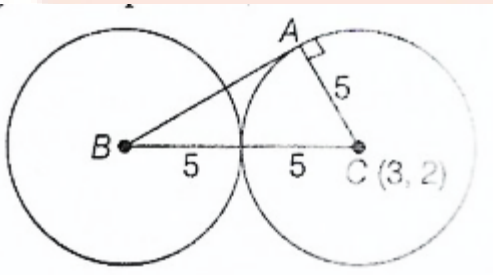
So

$$\begin{aligned} K &= \frac{6x_1 + 8y_1}{\lambda} \\ &= \frac{36 - 30\lambda + 64 + 40\lambda}{\lambda} \\ &= \frac{100}{\lambda} + 10 \\ &= 10 \pm 25\sqrt{2} \\ &= 5(2 \pm 5\sqrt{2}) \end{aligned}$$

51. Given expression of the circle is,

$$x^2 + y^2 - 6x - 4y - 12 = 0$$

$$(x - 3)^2 + (y - 2)^2 = 25$$



As triangle CAB is right angle triangle, so

$$\begin{aligned}AB^2 &= BC^2 - AC^2 \\&= 100 - 25 \\&= 75 \\&= 5\sqrt{3}\end{aligned}$$

52. Let the equation of the circle is,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

have center $(-g, -f)$. As it passes through the point $(3, 4)$

So,

$$9 + 16 + 6g + 8f + c = 0$$

The circle is intersecting another circle $(x^2 + y^2 = 36)$

orthogonally,

$$2g(0) + f(0) - c - 26$$

$$c = 36$$

From above calculations,

$$-6g - 8f = 61$$

By taking locus of the point $(-g, -f)$, we get $6x + 8y - 61 = 0$.

53. The equation of the circle is,

$$x^2 + y^2 - 8x - 2y + 8 = 0$$

$$(x^2 + y^2 - 8x - 2y + 8) + \lambda(x - 2) = 0$$

$$x^2 + y^2 + (\lambda - 8)x - 2y + (8 - 2\lambda) = 0$$

For minimum radius, it is necessary that center of circle lies on the line $x - 2 = 0$,

$$-\left(\frac{\lambda - 8}{2}\right) - 2 = 0$$

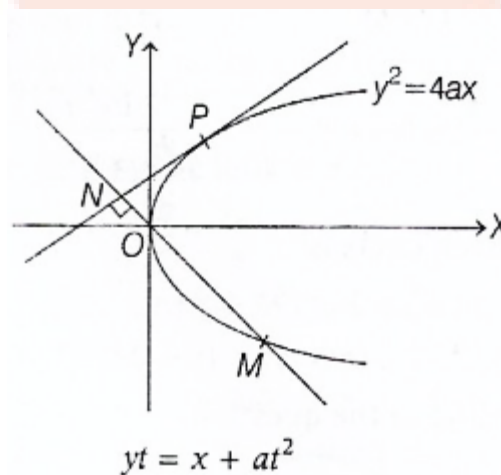
$$\lambda - 8 = -4$$

$$\lambda = 4$$

So, the equation of the required circle is,

$$x^2 + y^2 - 4x - 2y = 0$$

54. The point $P(at^2, 2at)$ is over the given parabola. So,



As NM is normal to the tangent passes through point O , so the equation of line NM is, $y = -tx$ and points,

$$N = \left(-\frac{at^2}{1+t^2}, \frac{at^3}{1+t^2} \right)$$

$$M = \left(\frac{4a}{t^2}, -\frac{4a}{t} \right)$$

So,

$$\begin{aligned} ON \cdot OM &= \sqrt{\frac{a^2 t^{40}}{(1+t^2)^2} + \frac{a^2 t^6}{(1+t^2)^2}} \times \sqrt{\frac{16a^2}{t^4} + \frac{16a^2}{t^2}} \\ &= \frac{4a^2 t^2}{t(1+t^2)} \sqrt{1+t^2} \sqrt{\frac{1}{t^2} + 1} \\ &= 4a^2 \end{aligned}$$

55. The ordinates of point A and B is α_1 and α_2 , so ordinate of point of intersection of the tangents is $\frac{\alpha_1 + \alpha_2}{2}$

So,

$$\alpha_3 = \frac{\alpha_1 + \alpha_2}{2}$$

$$\alpha_3 - \alpha_2 = \alpha_1 - \alpha_3$$

56. The equation of the ellipse is,

$$9x^2 + 4y^2 - 18x - 8y - 23 = 0$$

$$\frac{(x-1)^2}{4} + \frac{(y-1)^2}{9} = 1$$

Now, the equation of the latus rectum is,

$$y - 1 = \pm 3 \frac{\sqrt{5}}{3}$$

$$y = 1 \pm \sqrt{5}$$

57. The equation of the ellipse is,

$$4x^2 + 9y^2 = 36$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

The coordinate of end of the latus rectum lying in the second

quadrant is $P\left(-\sqrt{5}, \frac{4}{3}\right)$.

So, the equation of tangent at point P is,

$$-4\sqrt{5}x + 12y = 36$$

$$\sqrt{5}x - 3y + 9 = 0$$

58. The equation of the hyperbola is,

$$16x^2 - 25y^2 = 400$$

$$\frac{x^2}{25} - \frac{y^2}{16} = 1$$

And the asymptotes is,

$$y = \pm \frac{4}{5}x$$

Let point of hyperbola is represented by $A(5\sec\alpha, 4\tan\alpha)$. So the lengths from point A to asymptotes are,

So,

$$P = \frac{400}{41} \text{ and } \tan \frac{\theta}{2} = \frac{4}{5}$$

$$P \tan \frac{\theta}{2} = \frac{320}{41}$$

59. From given,

G_1 is the centroid of triangle BCD,

$$G_1 = \left(\frac{13}{3}, \frac{6}{3}, \frac{9}{3} \right)$$

G_2 is the centroid of triangle CDA,

$$G_2 = \left(\frac{12}{3}, \frac{7}{3}, \frac{7}{3} \right)$$

As the lines AG_1, BG_2, CG_3, DG_4 are concurrent, so point of concurrence of these four lines is point of intersection of lines AG_1 and BG_2 .

The equation of line AG_1 is,

$$\frac{x-3}{4/3} = \frac{y-2}{0} = \frac{z+1}{\frac{12}{3}} = r_1$$

So, the point on line AG_1 is $\left(3 + \frac{4}{3}r_1, 2, -1 + \frac{12}{3}r_1\right)$ and line BG_2

is

$$\frac{x-4}{0} = \frac{y-1}{\frac{4}{3}} = \frac{z-1}{\frac{4}{3}} = r_2$$

So, the point on line BG_2 is $\left(4, 1 + \frac{4}{3}r_2, 1 + \frac{4}{3}r_2\right)$

Let above point to be point of intersection,

$$3 + \frac{4}{3}r_1 = 4 \Rightarrow 2 = 1 + \frac{4}{3}r_2$$

$$\text{And } -1 + \frac{12}{3}r_1 = 1 + \frac{4}{3}r_2$$

From above values,

$$r_1 = \frac{3}{4}, r_2 = \frac{3}{4}$$

So, the required point of concurrence is $(4, 2, 2)$.

60. Given,

$$l + m + n = 0 \quad \dots\dots (I)$$

$$2lm + 2ln - mn = 0 \quad \dots\dots (II)$$

From equation (I) and (II),

$$2l(-l) - mn = 0$$

$$2l^2 + mn = 0 \quad \dots\dots (III)$$

Square both sides of equation (I),

$$l^2 + m^2 + n^2 + 2lm + 2mn + 2nl = 0$$

$$\text{As } l^2 + m^2 + n^2 = 1,$$

$$\text{So, } 2lm + 2mn + 2nl = -1 \quad \dots\dots (IV)$$

From equation (II) and (IV),

$$mn = \frac{-1}{3}$$

From equation (III),

$$6l^2 - 1 = 0$$

$$\text{So, } l_1 l_2 = -\frac{1}{6}$$

From equation (I) and (II),

$$2lm + (2l - m)(-l - m) = 0$$

$$2l^2 - lm - m^2 = 0$$

$$2\left(\frac{l}{m}\right)^2 - \left(\frac{l}{m}\right) - 1 = 0$$

$$\frac{l_1 l_2}{m_1 m_2} = \frac{-1}{2}$$

$$\frac{l_1 l_2}{\left(\frac{-1}{6}\right)} = \frac{m_1 m_2}{\left(\frac{2}{6}\right)}$$

Similarly,

$$\frac{l_1 l_2}{\left(\frac{-1}{6}\right)} = \frac{n_1 n_2}{\left(\frac{2}{6}\right)}$$

So,

$$\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$$

$$\cos \theta = \frac{3}{6}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

61. Let the points $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$. So, the point of intersection of planes P_1 , P_2 , and P_3 is $P(a, b, c)$.

Now,

The equation of the plane ABC is,

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Above plane passes through point (α, β, γ) , so

$$\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1$$

Take locus of point (a, b, c) ,

$$\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 1$$

62. Solve as follow,

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \cos 2x}{\cos 3x (3^{\sin 2x} - 1)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{2 \cos^2 x}{\frac{\cos 3x}{\sin 3x} \times (3^{\sin 2x} - 1)}$$

$$\text{Let } x = \frac{\pi}{2} - h,$$

So,

$$\begin{aligned}
 \lim_{h \rightarrow 0} \frac{2 \sin^2 h}{\frac{\sin 3h}{\cos 3h} (3^{\sin 2h} - 1)} &= \lim_{h \rightarrow 0} \frac{2 \frac{\sin^2 h}{h^2} \times h^2 (\cos 3h)}{\left[\left(\frac{\sin 3h}{\cos 3h} \times 3h \right) \left(\frac{3^{\sin 2h} - 1}{\sin 2h} \right) \left(\frac{\sin 2h}{2h} \right) \right]} \\
 &= \frac{2(1)}{(1 \times 3)[\log_e 3](2)} \\
 &= \frac{1}{3 \log_e 3}
 \end{aligned}$$

63. Solve as follow,

$$\begin{aligned}
 P &= \lim_{n \rightarrow \infty} n^{-nk} \left\{ (n+1) \left(n + \frac{1}{2} \right) \left(n + \frac{1}{2^2} \right) \dots \left(n + \frac{1}{2^{k-1}} \right) \right\}^n \\
 &= \lim_{n \rightarrow \infty} \left[\frac{(n+1) \left(n + \frac{1}{2} \right) \left(n + \frac{1}{2^2} \right) \dots \left(n + \frac{1}{2^{k-1}} \right)}{n^k} \right]^n \\
 &= \lim_{n \rightarrow \infty} \left[\left(\frac{n + \frac{1}{2^0}}{n} \right) \left(\frac{n + \frac{1}{2^2}}{n} \right) \dots \left(\frac{n + \frac{1}{2^{k-1}}}{n} \right) \right]^n
 \end{aligned}$$

So,

$$\begin{aligned}
 \log P &= n \left[\log \left(\frac{n + \frac{1}{2^0}}{n} \right) + \log \left(\frac{n + \frac{1}{2^2}}{n} \right) + \dots + \log \left(\frac{n + \frac{1}{2^{k-1}}}{n} \right) \right] \\
 &= \left[\frac{\log \left(1 + \frac{1}{2^0 n} \right)}{1/n} + \frac{\log \left(1 + \frac{1}{2^2 n} \right)}{1/n} + \dots + \frac{\log \left(1 + \frac{1}{2^{k-1} n} \right)}{1/n} \right] \\
 &= \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \dots + \frac{1}{2^{k-1}} \\
 &= 2 \left(1 - \frac{1}{2^k} \right) \\
 P &= e^{2 \left(1 - \frac{1}{2^k} \right)}
 \end{aligned}$$

64. At $x = 1$,

$$\text{LHL} = 5 \text{ and } \text{RHL} = -1$$

So, the function is discontinuous at $x = 1$,

At $x = 2$,

$$\text{LHL} = 2 \text{ and } \text{RHL} = 1$$

So, the function is discontinuous at $x = 2$.

As $a > b$,

$$a = 2,$$

$$b = 1$$

So,

$$3a - b = 5$$

65. Let $x = \cos 2\theta$,

So,

$$\begin{aligned}\frac{1-x}{1+x} &= \frac{1-\cos 2\theta}{1+\cos 2\theta} \\ &= \tan^2 \theta\end{aligned}$$

So the function can be written as,

$$\begin{aligned}f(x) &= \cos^2 \theta \\ &= \frac{1+\cos 2\theta}{2} \\ &= \frac{1+x}{2}\end{aligned}$$

So,

$$f'(x) = \frac{1}{2}$$

66. The functional relation is,

$$f(xy+1) = f(x)f(y) - f(y) - x + 2$$

Take $y = 0$,

$$\begin{aligned}f(1) &= f(x) \cdot f(0) - f(0) - x + 2 \\ &= f(x) - 1 - x + 2 \quad (\text{As, } f(0) = 1)\end{aligned}$$

Differentiate above equation with respect to x .

$$0 = \frac{df(x)}{dx} - 0 - 1 + 0$$

$$\frac{df(x)}{dx} = 1 \quad (\text{At, } x = e)$$

67. From the equation,

$$\begin{aligned} y &= \sin(\log(x^2 + 2x + 1)) \\ &= \sin[2\log(x + 1)] \end{aligned}$$

Differentiate with respect to x .

$$\frac{dy}{dx} = [\cos(2\log(x + 1))] \times \left(\frac{2}{x + 1}\right)$$

$$(x + 1) \frac{dy}{dx} = 2 \cos(2(\log(x + 1)))$$

Differentiate above equation with respect to x ,

$$(x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = -2(\sin(2\log(1 + x))) \left(\frac{2}{x + 1}\right)$$

$$(x + 1)^2 \frac{d^2y}{dx^2} + (x + 1) \frac{dy}{dx} = -4y$$

68. Points of intersection of the given curves other than origin is $(4, 4)$.

The slope of curve (1),

$$\begin{aligned} m_1 &= \left. \frac{dy}{dx} \right|_{(4,4)} \\ &= \left. \frac{2x}{4} \right|_{(4,4)} \\ &= 2 \end{aligned}$$

The slope of curve (2),

$$\begin{aligned} m_2 &= \left. \frac{dy}{dx} \right|_{(4,4)} \\ &= \left. \frac{4}{2y} \right|_{(4,4)} \\ &= \frac{1}{2} \end{aligned}$$

So, the angle between the tangents is,

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{2 - \frac{1}{2}}{1 + 2 \left(\frac{1}{2} \right)} \right| \\ &= \frac{3}{4} \end{aligned}$$

$$\text{So, } \theta = \tan^{-1} \left(\frac{3}{4} \right)$$

$$\theta = \sin^{-1} \frac{3}{5}$$

69. For $x > 0$,

$$f(x) = \frac{x}{1+x} - \log(1+x)$$

Now,

$$\begin{aligned} f'(x) &= \frac{(x+1) - x}{(1+x)^2} - \frac{1}{(1+x)} \\ &= \frac{1}{(1+x)^2} - \frac{1}{1+x} \\ &= \frac{1 - (1+x)}{(1+x)^2} \\ &= \frac{-x}{(1+x)^2} < 0 \end{aligned}$$

As $f'(x) < 0, \forall x > 0$,

The function $f(x)$ is a strictly decreasing function for $x > 0$.

$$\text{Now, } f(0^+) = 0$$

$$\text{So, } f(x) < 0$$

70. From Lagrange's Mean value theorem,

As $y = x^3$ is continuous and differentiable function

So, there exists $x = c$,

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)}$$

$$3c^2 = \frac{8 - (-1)}{3}$$

$$c^2 = 1$$

$$c = \pm 1$$

Thus, the point is $(1, 1)$.

71. Let the speed of the motor boat relative to water is $V > C$. The speed of the motor boat relative to the land is, $V - C$.

The distance covered by boat is S in $\frac{S}{V - C}$ h time and the consumption is,

$$f(v) = kv^3 \cdot \frac{S}{V - C'}.$$

Here, k is constant.

For minimum consumption,

$$f'(v) = 0$$

$$kS \frac{(V - C)3V^2 - V^3(1)}{(V - C)^2} = 0$$

$$3V^3 - 3V^2C - V^3 = 0$$

$$2V^3 - 3V^2C = 0$$

$$V = \frac{3C}{2}$$

72. For $x < 1$,

$$\begin{aligned} I &= \int \frac{x - x^2}{\sqrt{1 - x}} dx \\ &= \int \frac{x(1 - x)}{\sqrt{1 - x}} dx \\ &= \int x\sqrt{1 - x} dx \end{aligned}$$

Consider $t^2 = 1 - x$

So, $dx = -2t dt$

Now,

$$\begin{aligned} I &= \int (1 - t^2)t(-2t) dt \\ &= 2 \int (t^4 - t^2) dt \\ &= 2 \left[\frac{t^5}{5} - \frac{t^3}{3} \right] + c \end{aligned}$$

$$\begin{aligned} I &= \frac{2}{15} t^3 (3t^2 - 5) + c \\ &= \frac{2}{15} (1-x)^{3/2} [3(1-x) - 5] + c \\ &= \frac{-2}{15} (1-x)^{3/2} (3x+2) + c \end{aligned}$$

73. Solve as follow,

$$\begin{aligned} I &= \int \frac{dx}{(2 \sin x + \sec x)^4} \\ &= \int \frac{\sec^4 x}{(2 \tan x + \sec^2 x)^4} dx \\ &= \int \frac{(1 + \tan^2 x)}{(1 + \tan x)^8} \sec^2 x dx \end{aligned}$$

Let $\tan x = t$

So,

$$\sec^2 x dx = dt$$

So,

$$\begin{aligned}
 I &= \int \frac{1+t^2}{(1+t)^8} dt \\
 &= \int \frac{(1+t)^2 - 2t}{(1+t)^8} dt \\
 &= \int \frac{dt}{(1+t)^6} - 2 \int \frac{dt}{(1+t)^7} + 2 \int \frac{dt}{(1+t)^8} \\
 &= -\frac{1}{5}(1+\tan x)^{-5} + \frac{1}{3}(1+\tan x)^{-6} - \frac{2}{7}(1+\tan x)^{-7} + k
 \end{aligned}$$

So, $A = -\frac{1}{5}$, $B = \frac{1}{3}$ and $C = -\frac{2}{7}$

$$\Rightarrow A + B + C = -\frac{16}{105}$$

74. From given expression,

$$\int \frac{2x^2 - 1 + x^2 \sqrt{x^2 + 4}}{x^2 (x^2 + 4)} dx = \frac{9}{4}$$

So,

$$\begin{aligned}
 &= \int \frac{dx}{x^2 + 4} - \frac{1}{4} \int \frac{dx}{x^2} + \int \frac{dx}{\sqrt{x^2 + 4}} \\
 &= \frac{9}{8} \tan^{-1} \left(\frac{x}{2} \right) + \frac{1}{4x} + \sinh^{-1} \left(\frac{x}{2} \right) + c
 \end{aligned}$$

75. From the given data,

$$I_n = \int_0^{\pi/4} \tan^n x dx, n \geq 2$$

Now,

$$\begin{aligned} F_n &= I_n + I_{n-2} \\ &= \int_0^{\pi/4} (\tan^n x + \tan^{n-2} x) dx \\ &= \int_0^{\pi/4} (\tan^2 x + 1) \tan^{n-2} x dx \\ &= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x dx \end{aligned}$$

Let $t = \tan x$,

$$\sec^2 x dx = dt$$

So,

$$x = \frac{\pi}{4}, t = 1$$

So,

$$\begin{aligned} F_n &= \int_0^1 t^{n-2} dt \\ &= \frac{t^{n-1}}{n-1} \Big|_0^1 \\ &= \frac{1}{n-1} \end{aligned}$$

Same as,

$$\begin{aligned}
 F_{n+1} &= \frac{1}{n} \\
 &= \frac{n - (n-1)}{n(n-1)} \\
 &= \frac{1}{n(n-1)}
 \end{aligned}$$

76. Solve the expression,

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} n \left[\frac{1}{3n^2 + 8n + 4} + \frac{1}{3n^2 + 16n + 16} + \dots + \frac{1}{15n^2} \right] \\
 &= \lim_{n \rightarrow \infty} n \sum_{r=1}^n \frac{1}{3n^2 + 8nr + (2r)^2} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{3n^2 + 8\left(\frac{r}{n}\right) + 4\left(\frac{r}{n}\right)^2} \\
 &= \int_0^1 \frac{dx}{2 + 8x + 4x^2} \\
 &= \frac{1}{4} \times \frac{1}{2(1/2)} \left[\ln \left(\frac{(x+1) - \frac{1}{2}}{(x+1) + \frac{1}{2}} \right) \right]_0^1 \\
 &= \frac{1}{4} \ln \frac{9}{5}
 \end{aligned}$$

77. From given expression,

$$I = \int_0^3 |x^2 - 3x + 2| dx$$

As,

$$x^2 - 3x + 1 = (x - 2)(x - 1)$$

$$x^2 - 3x + 1 < 0, x \in (1, 2)$$

And

$$x^2 - 3x + 2 \geq 0, x \in R - (1, 2)$$

So,

$$\begin{aligned} I &= \left[\int_0^1 (x^2 - 3x + 2) dx - \int_1^2 (x^2 - 3x + 2) dx + \int_2^3 (x^2 - 3x + 2) dx \right] \\ &= \left[\left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_0^1 - \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_1^2 \right. \\ &\quad \left. + \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_2^3 \right] \\ &= \left[\left(\frac{1}{3} - \frac{3}{2} + 2 \right) - \left[\left(\frac{8}{3} - \frac{12}{2} + 4 \right) - \left(\frac{1}{3} - \frac{3}{2} + 2 \right) \right] \right] \\ &\quad + \left[\left(\frac{27}{3} - \frac{27}{2} + 6 \right) - \left(\frac{8}{3} - \frac{12}{2} + 4 \right) \right] \\ &= \frac{11}{6} \end{aligned}$$

78. As,

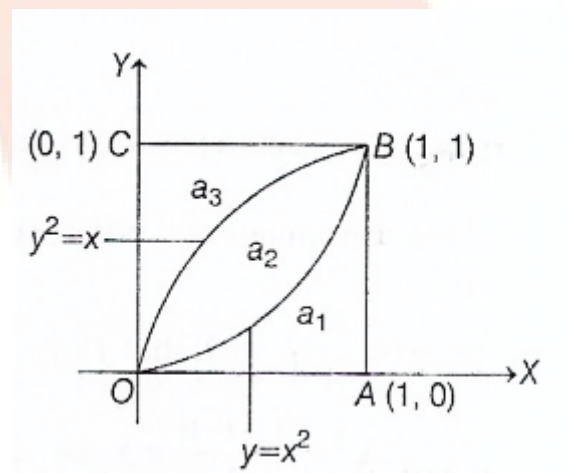
$$a_1 + a_2 + a_3 = 1 \quad \dots\dots (I)$$

And $a_1 = a_3$ (symmetry)

Now,

$$\begin{aligned} a_2 &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left[\frac{2}{3} x^{3/2} - \frac{x^3}{3} \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{3} \\ &= \frac{1}{3} \end{aligned}$$

Consider the figure,



From figure,

$$a_1 + a_3 + \frac{1}{3} = 1$$

$$a_1 + a_3 = \frac{2}{3} \quad \dots\dots (II)$$

From symmetry and equation (II),

$$\begin{aligned} a_1 &= a_3 \\ &= \frac{1}{3} \\ &= a_2 \end{aligned}$$

So,

$$\begin{aligned} a_1 + 2a_2 + 3a_3 &= \frac{1}{3} + \frac{2}{3} + \frac{3}{3} \\ &= 2 \end{aligned}$$

79. From parabola equation,

$$y^2 = 4a(x + a) \quad \dots\dots (I)$$

$$2y \frac{dy}{dx} = 4a$$

$$a = \frac{1}{2} y \frac{dy}{dx} \quad \dots\dots (II)$$

From equation (I) and (II),

$$y^2 = 4x \left(\frac{1}{2} y \frac{dy}{dx} \right) + 4 \left(\frac{1}{2} y \frac{dy}{dx} \right)^2$$

$$y \left(\frac{dy}{dx} \right)^2 + 2x \frac{dy}{dx} - y = 0$$

80. From given differential equation,

$$\frac{dy}{dx} + y(\tan x) = 2x + x^2 \tan x$$

The integrating factor is,

$$\begin{aligned} \text{I.F.} &= e^{\int \tan x dx} \\ &= \sec x \end{aligned}$$

Solution of the differential equation is,

$$\begin{aligned} y(\sec x) &= \int (2x + x^2 \tan x) \sec x dx \\ &= \int (2x) \sec x dx + \int (x^2 \tan x) \sec x dx \\ &= \int 2x \sec x dx + x^2 \sec x - \int 2x \sec x dx \\ y &= x^2 + c \cos x \end{aligned}$$

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Solutions – Physics

81. Given info:

Unit of mass is A kg, unit of length is B m and unit of time is C s.

Use the below represented equilibrium equation.

$$N_1 u_1 = N_2 u_2$$

$$10 \text{ kg} \cdot \text{m/s}^2 = N_2 \left(\frac{AB}{C^2} \right)$$

$$N_2 = 10 A^{-1} B^{-1} C^2$$

82. During the deceleration, if a body is slowing down then the acceleration and velocity of the body are directed opposite to each other. Thus the angle between acceleration and velocity may be 180° and 0° .

83. Given info:

The initial velocity of a particle is $(\hat{i} + 2\hat{j}) \text{ ms}^{-1}$.

Here, the initial velocity of a particle in x direction is $u_x = 1 \text{ ms}^{-1}$ and the initial velocity of a particle in y direction is $u_y = 2 \text{ ms}^{-1}$.

So, the acceleration of a particle in x direction is $a_x = 0$ and the acceleration of a particle in y direction is $a_y = -10 \text{ ms}^{-2}$.

The horizontal distance covered by projectile in time t is,

$$\begin{aligned}x &= u_x t \\&= (1 \text{ ms}^{-1})t \\&= t\end{aligned}$$

The vertical distance covered by projectile in time t is,

$$\begin{aligned}y &= u_y t + \frac{1}{2} a_y t^2 \\&= (2 \text{ ms}^{-1})t + \frac{1}{2}(-10 \text{ ms}^{-2})t^2 \\&= 2t - 5t^2 \\y &= 2x - 5x^2\end{aligned}$$

84. Write the expression of the trajectory of projectile.

$$\begin{aligned}y &= x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} \\&= x \tan(45^\circ) - \frac{gx^2}{2u^2 \cos^2(45^\circ)} \\&= x - \frac{gx^2}{u^2} \quad \dots (1)\end{aligned}$$

Substitute $x = 4$ and $y = 3$ in Equation (1).

$$3 = 4 - \frac{g(4)^2}{u^2}$$

$$\frac{g(4)^2}{u^2} = 1$$

$$u^2 = 16g$$

From Equation (1),

$$\begin{aligned}y &= x - \frac{gx^2}{16g} \\&= x - \frac{x^2}{16} \quad \dots (2)\end{aligned}$$

For the maximum value of x , the value of $y = 0$ substitute in (2).

$$0 = x - \frac{x^2}{16}$$

$$x\left(1 - \frac{x}{16}\right) = 0$$

$$x = 0 \text{ and } x = 16$$

85. The displacement of a hanging block is given by,

$$s = ut + \frac{1}{2}at^2$$

Here, displacement is 2 m.

$$2 = 0(t) + \frac{1}{2}a(2)^2$$

$$2 = 2a$$

$$a = 1 \text{ ms}^{-2}$$

So, the velocity of a 10 kg block after 2 s is equal to the velocity of a hanging block.

$$v = u + at$$

$$= 0 + 1 \times 2$$

$$= 2 \text{ ms}^{-1}$$

By using work energy theorem the total work done by a hanging block due to gravity is equal to the kinetic energy of a 10 kg block.

$$\frac{1}{2}(10)(2)^2 = m(10)(2)$$

$$m = 1 \text{ kg}$$

So, the weight of a hanging block is,

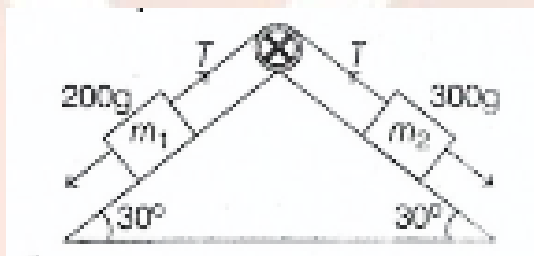
$$W_{\text{block}} = (1 \text{ kg}) \times (10 \text{ ms}^{-2})$$

$$= (10 \text{ kg} \cdot \text{m/s}^2)$$

$$= 10 \text{ N}$$

$$\approx 11.11 \text{ N}$$

86. The following figure represents the required free body diagram of the system.



From figure,

$$T - m_1 g \sin \theta = m_1 a \quad \text{..... (1)}$$

And,

$$m_2 g \sin \theta - T = m_2 a \quad \text{..... (2)}$$

Add Equation (1) and (2).

$$T - m_1 g \sin \theta + m_2 g \sin \theta - T = m_1 a + m_2 a$$

$$g \sin \theta (m_2 - m_1) = a (m_1 + m_2)$$

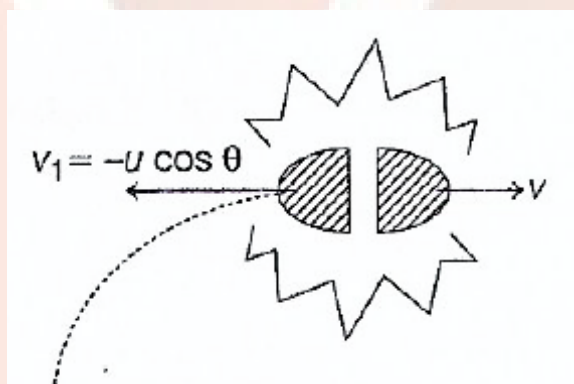
$$a = \frac{g \sin \theta (m_2 - m_1)}{(m_1 + m_2)}$$

$$a = \frac{g \sin 30^\circ (300 - 200)}{(300 + 200)}$$

$$a = \frac{1}{10} g$$

$$a = 10\% \text{ of } g$$

87. The following figure represents the required diagram.



Before explosion momentum is given by,

$$M_{\text{be}} = mu \cos \theta \times 2$$

After explosion momentum is given by,

$$M_{\text{ae}} = -mu \cos \theta + mv_2$$

Use conservation of momentum.

$$mu \cos \theta \times 2 = -mu \cos \theta + mv_2$$

$$mv_2 = 3mu \cos \theta$$

$$v_2 = 3u \cos \theta$$

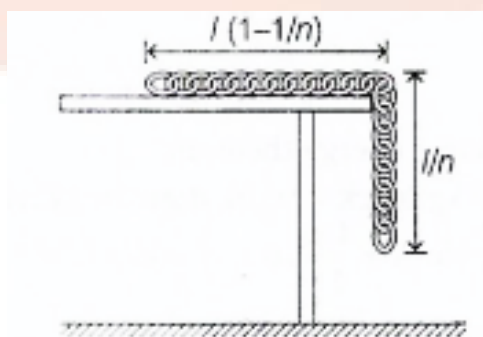
The kinetic energy of first part is given by,

$$\begin{aligned} E_1 &= \frac{1}{2}mv_1^2 \\ &= \frac{1}{2}m(-u \cos \theta)^2 \\ &= \frac{1}{2}mu^2 \cos^2 \theta \end{aligned}$$

The kinetic energy of second part is given by,

$$\begin{aligned} E_2 &= \frac{1}{2}mv_2^2 \\ &= \frac{1}{2}m(3u \cos \theta)^2 \\ &= \frac{9}{2}mu^2 \cos^2 \theta \\ &= 9E_1 \end{aligned}$$

88. The following figure represents the required diagram.



With $\frac{1}{n}$ th part of hanging the potential energy of chain is given by,

$$P_{\frac{1}{n}\text{th}} = -\frac{Mgl}{2n^2}$$

When the chain leaves table, the potential energy of chain is given by,

$$P_L = -\frac{Mgl}{2}$$

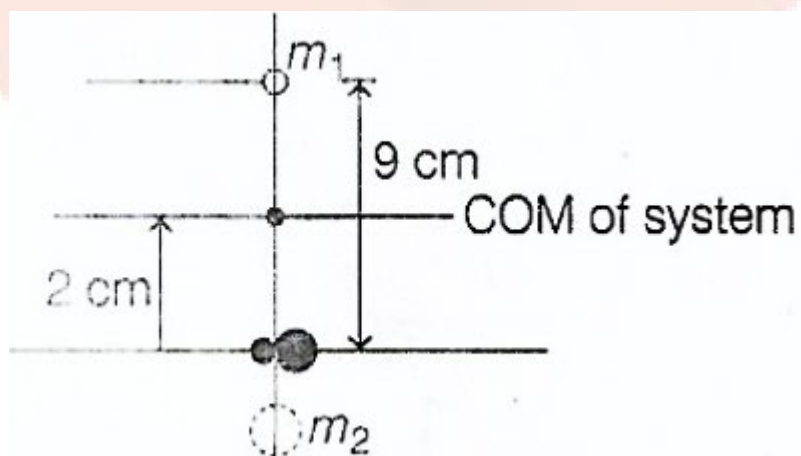
From the conservation of energy,

Kinetic energy = Loss in potential energy

$$\frac{1}{2}Mv^2 = \frac{Mgl}{2}\left(1 - \frac{1}{n^2}\right)$$

$$v = \sqrt{gl\left(1 - \frac{1}{n^2}\right)}$$

89. The below figure represent the required diagram.



Consider, at initial time particle is present at origin and the masses of the bodies are m and $2m$.

So,

$$2 = \frac{m \times 9 + 2m \times y_2}{3m}$$

$$2 = \frac{9 + 2y_2}{3}$$

$$y_2 = -1.5 \text{ cm}$$

90. The conservation is given if the body rolls down in an inclined plane as,

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(mk^2)\left(\frac{v}{R}\right)^2$$

$$mgh = \frac{1}{2}mv^2\left(1 + \frac{k^2}{R^2}\right)$$

$$v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

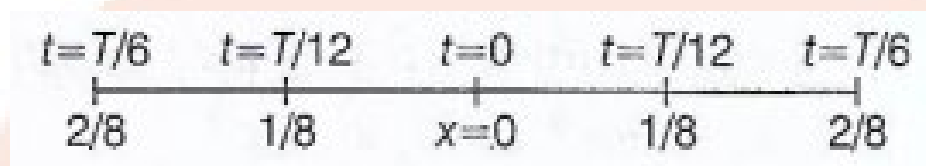
For a sphere, $\frac{k^2}{R^2} = \frac{2}{5}$ and for a ring $\frac{k^2}{R^2} = 1$

So, the ratio is,

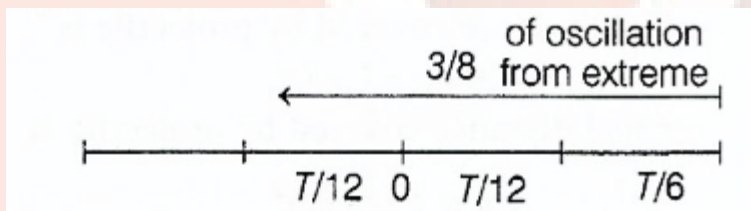
$$\frac{v_{\text{ring}}^2}{v_{\text{sphere}}^2} = \frac{1 + \frac{2}{5}}{1 + 1}$$

$$= 0.7$$

91. The below figure represent the required diagram.



For an oscillation from an extreme of a displacement of $\frac{3}{8}$ th is shown below,



So, the time is,

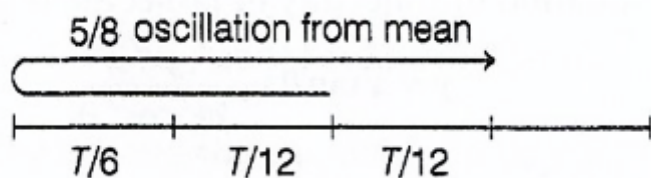
$$\text{Time} = \frac{T}{6} + \frac{T}{12} + \frac{T}{12}$$

$$= \frac{(2+1+1)T}{12}$$

$$x = \frac{T}{3}$$

$$T = 3x$$

Now, for $\frac{5}{8}$ th of oscillation from mean is shown below,



So, the time is,

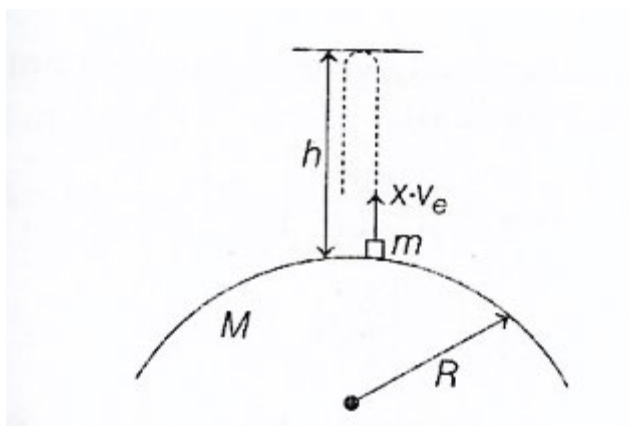
$$\begin{aligned}
 \text{Time} &= \frac{T}{12} + \frac{T}{6} + \frac{T}{6} + \frac{T}{12} + \frac{T}{12} \\
 &= \frac{(1+2+2+1+1)T}{12} \\
 &= \frac{7(3x)}{12} \\
 &= \frac{7}{4}x
 \end{aligned}$$

92. The escape velocity is given by,

$$V_{\text{Escape}} = \sqrt{\frac{2GM}{R}}$$

Consider, if the object is thrown with a speed of x then the object rises up to height of h , then by using the conservation of energy we obtain.

The below figure represent the required diagram.



$$\frac{1}{2}m(xV_{\text{Escape}})^2 - \frac{GMm}{R} = 0 - \frac{GMm}{R+h}$$

$$\frac{1}{2}m(xV_{\text{Escape}})^2 = \frac{GMm}{R} - \frac{GMm}{R+h}$$

$$\frac{1}{2}x^2 \left(\sqrt{\frac{2GM}{R}} \right)^2 = \frac{GM}{R} - \frac{GM}{R+h}$$

$$\frac{1}{2}x^2 \frac{2GM}{R} = \frac{GM}{R} - \frac{GM}{R+h}$$

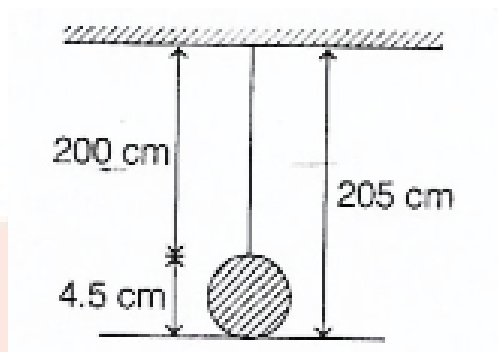
$$\frac{x^2}{R} = \frac{1}{R} - \frac{1}{R+h}$$

$$x^2 = 1 - \frac{R}{R+h}$$

$$h = \frac{Rx^2}{1-x^2}$$

Thus no option is correct.

93. The below figure represent the required diagram.



Due to centrifugal force and load the elongation in wire is given by,

$$\Delta l = (205 - 204.5) \text{ cm}$$

$$= 0.5 \text{ cm}$$

At lowest if the velocity of sphere is v , then

$$Y = \frac{\left(Mg + \frac{Mv^2}{R} \right) L}{A \Delta l}$$

$$Mg + \frac{Mv^2}{R} = \frac{YA \Delta l}{L}$$

Here, $R = 202.25 \text{ cm}$

So,

$$2 \times 10 + \frac{2v^2}{(202.25 \text{ cm}) \times 10^{-2}} = \frac{(2 \times 10^{11})(0.24 \times 10^{-6})(0.5 \times 10^{-2})}{2}$$

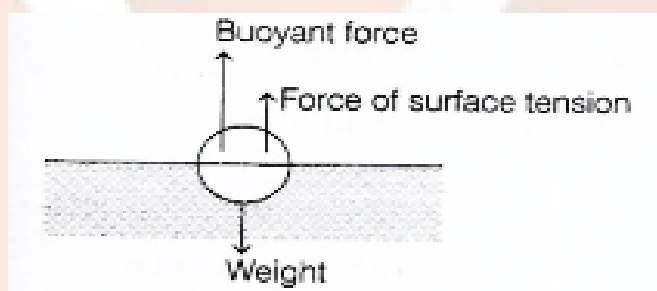
$$2 \times 10 + \frac{2v^2}{(202.25 \text{ cm}) \times 10^{-2}} = 0.12 \times 10^3$$

$$\frac{2v^2}{(202.25 \text{ cm}) \times 10^{-2}} = 100$$

$$v^2 = 101.125$$

$$v = 10.05 \text{ ms}^{-1} \text{ or } v = 10 \text{ ms}^{-1}$$

94. The following figure represents the required diagram.



Write the expression of the weight of the body by using the diagram.

Weight of the body = Buoyant force + Force due to surface tension

$$\frac{4}{3}\pi r^3 \rho g = \frac{2}{3}\pi r^3 dg + 2\pi r\sigma$$

$$\frac{2}{3}\pi r^3 (2\rho - d) = 2\pi r\sigma$$

$$r^2 = \frac{3\sigma}{g(2\rho - d)}$$

$$r = \sqrt{\frac{3\sigma}{g(2\rho - d)}}$$

95. Write the expression of the change in length due to change of temperature by t is given by,

$$\begin{aligned}\Delta DC^2 &= \Delta AC^2 - \Delta AD^2 \\ &= \{l(1 + \alpha_2 t)\}^2 - \left\{\frac{l}{2}(1 + \alpha_1 t)\right\}^2 \\ &= l^2(2\alpha_2 t) - \frac{l^2}{4}(2\alpha_1 t)\end{aligned}$$

Substitute change in length is equal to zero.

$$0 = l^2(2\alpha_2 t) - \frac{l^2}{4}(2\alpha_1 t)$$

$$\alpha_1 = 4\alpha_2$$

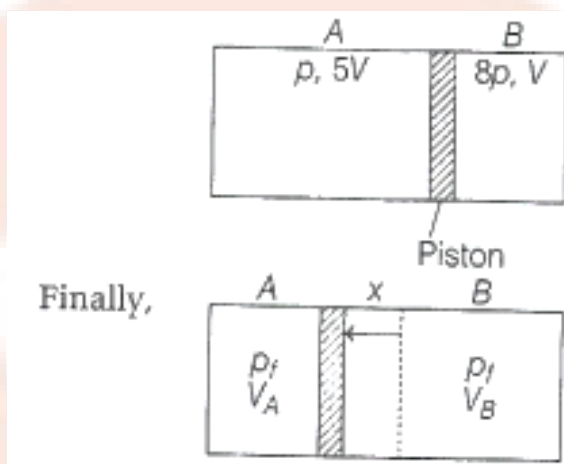
96. If the ice melts, the density of ice increases or volume of ice decreases.

When the water changes to steam, its volume increases.

If the pressure increases, the melting point of ice decreases.

The boiling of water increases, if the pressure is increases.

97. The below figure represent the required diagram.



Write the expression of adiabatic expansion in part A.

$$p(5V)^r = p_f V_A^r \quad \text{..... (i)}$$

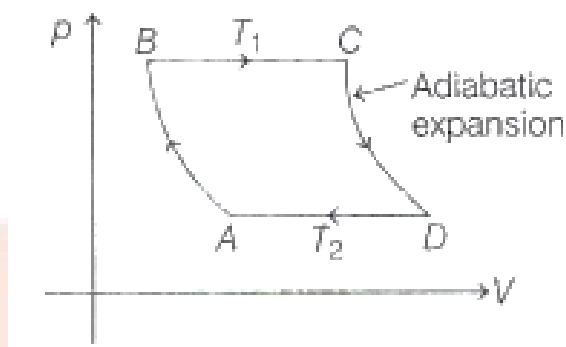
Write the expression of adiabatic expansion in part B.

$$8p(V)^r = p_f V_B^r \quad \text{..... (i)}$$

By Substituting $r = 1.5$ in we get,

$$V_A = \frac{10}{3}V$$

98. The below figure represent the required diagram,



Given information:

Volume at C is equal to V and the volume at D is equal to $32V$.

Write the expression of adiabatic expansion.

$$T_C V_C^{\gamma-1} = T_D V_D^{\gamma-1}$$

$$\frac{T_C}{T_D} = \left(\frac{V_D}{V_C} \right)^{\gamma-1}$$

$$\frac{T_C}{T_D} = \left(\frac{32V}{V} \right)^{\frac{7}{5}-1}$$

$$\frac{T_C}{T_D} = 32^{\frac{2}{5}}$$

$$\frac{T_C}{T_D} = 4$$

So, the efficiency of Carnot cycle is given by,

$$\begin{aligned}\eta &= 1 - \frac{T_D}{T_C} \\ &= 1 - \frac{1}{4} \\ &= 0.75\end{aligned}$$

99. The RMS velocity of hydrogen is given by,

$$\begin{aligned}v_{\text{rms(Hydrogen)}} &= \sqrt{\frac{3RT}{M}} \\ &= \sqrt{\frac{3kT}{m}}\end{aligned}$$

The escape velocity of moon is given by,

$$v_{\text{escape(moon)}} = \sqrt{2gR}$$

Here, the RMS velocity is equal to escape velocity of moon.

$$\begin{aligned}\sqrt{\frac{3kT}{m}} &= \sqrt{2gR} \\ \frac{3kT}{m} &= 2gR \\ T &= \frac{2mgR}{3k}\end{aligned}$$

100. The initial fundamental frequency is given by,

$$n_1 = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$= \frac{1}{2l} \sqrt{\frac{Vg(2000)}{\mu}}$$

The final fundamental frequency is given by,

$$n_2 = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$= \frac{1}{2l} \sqrt{\frac{Vg\left(2000 - \frac{1000}{2}\right)}{\mu}}$$

$$= \frac{1}{2l} \sqrt{\frac{1500Vg}{\mu}}$$

So,

$$\frac{n_1}{n_2} = \sqrt{\frac{2000}{1500}}$$

$$\frac{n_1}{n_2} = \sqrt{\frac{4}{3}}$$

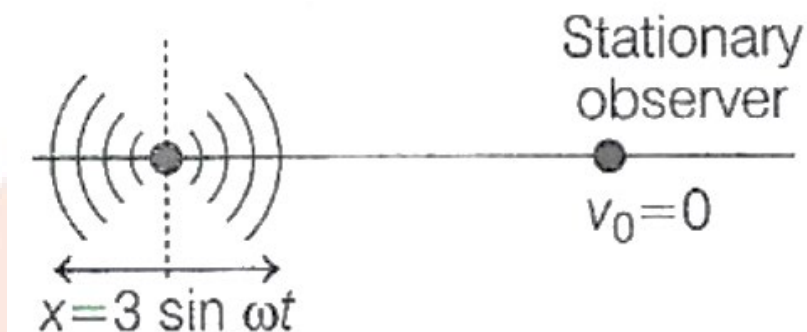
$$n_2 = \frac{n_1 \sqrt{3}}{2}$$

$$n_2 = \frac{(200 \text{ Hz}) \sqrt{3}}{2}$$

$$n_2 = 173.2 \text{ Hz}$$

101. The frequency of source is $f = 120$ Hz.

The following figure represents the required diagram.



Write the expression of the instantaneous speed of the source.

$$v = \frac{dx}{dt}$$

$$= 3\omega \sin \omega t$$

Here, the difference between the maximum and minimum frequency is 22 Hz.

$$f_{\max} - f_{\min} = f \left(\frac{v}{v - v_s} \right) - f \left(\frac{v}{v + v_s} \right)$$

$$f_{\max} - f_{\min} = f \left(\frac{v}{v - v_s} - \frac{v}{v + v_s} \right)$$

$$f \left(\frac{v^2 + vv_s - v^2 + vv_s}{v^2 - v_s^2} \right) = f_{\max} - f_{\min}$$

$$120 \left(\frac{2vv_s}{v^2 - v_s^2} \right) = 22$$

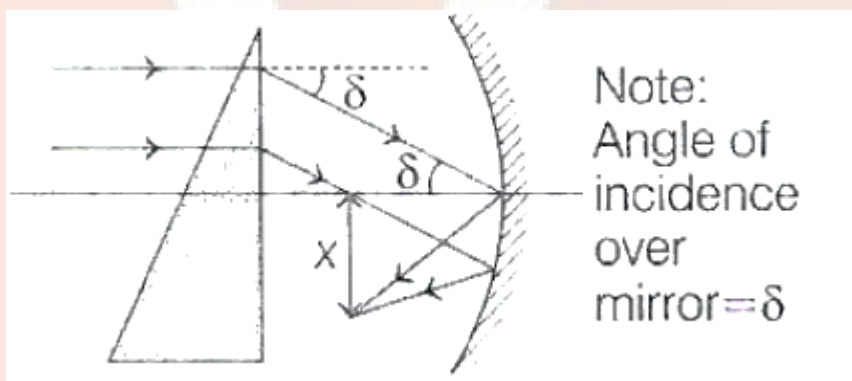
$$\frac{2(330)(3\omega)}{(330)^2 - (3\omega)^2} = \frac{22}{120}$$

$$\omega = 10 \text{ rad} \cdot \text{s}^{-1}$$

102. The parallel beam is deviated by prism by angle,

$$\begin{aligned} \delta &= (\mu - 1)A \\ &= (1.5 - 1)(1.8^\circ) \\ &= (0.9^\circ) \left(\frac{\pi}{180^\circ} \right) \text{radian} \end{aligned}$$

This beam from prism is also parallel and is conserved at a point in focal plane by concave mirror.



So, the distance is,

$$\begin{aligned} x &= F \times \delta \\ &= \frac{R}{2} \times \delta \\ &= \frac{20}{2} (0.9^\circ) \left(\frac{\pi}{180^\circ} \right) \end{aligned}$$

$$x = 0.157 \text{ cm}$$

$$= 1.57 \text{ mm}$$

103. For minimum separation,

$$d_{\min} = \frac{1.22 f \lambda}{D}$$

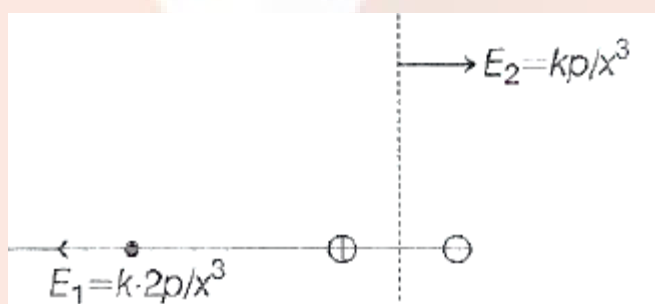
$$= \frac{1.22 \times (5 \text{ cm}) (5500 \text{ \AA})}{(8 \text{ mm})}$$

$$= \frac{1.22 \times (5 \times 10^{-2}) (5500 \times 10^{-10})}{(8 \times 10^{-3})}$$

$$= 41.9 \times 10^{-7} \text{ m}$$

$$\approx 4.2 \text{ \mu m}$$

104. The below figure represents the required diagram.



The magnitude of electric field on axis is given by,

$$E_1 = \frac{2kp}{r^3}$$

The magnitude of electric field on equatorial axis is given by,

$$E_2 = \frac{kp}{(2r)^3}$$
$$= \frac{kp}{8r^3}$$

So,

$$E_1 = 16E_2$$

So, $x = 16$

105. Write the expression of electric field at A, in the vector form.

$$E_A = \left\{ \frac{kp}{(1^2 + 2^2 + 3^2)^{3/2}} \right\} (\hat{i} + 2\hat{j} + 3\hat{k})$$
$$= \frac{kp}{(14)^{3/2}} (\hat{i} + 2\hat{j} + 3\hat{k})$$

Write the expression of electric field at B, in the vector form.

$$E_B = \left\{ \frac{kp}{(1^2 + 1^2 + (-1)^2)^{3/2}} \right\} (\hat{i} + \hat{j} - \hat{k})$$
$$= \frac{kp}{(3)^{3/2}} (\hat{i} + \hat{j} - \hat{k})$$

Write the expression of electric field at C, in the vector form.

$$E_C = \left\{ \frac{kp}{(2^2 + 2^2 + 2^2)^{3/2}} \right\} (2\hat{i} + 2\hat{j} + 2\hat{k})$$

$$= \frac{kp}{(12)^{3/2}} (\hat{i} + 2\hat{j} + 3\hat{k})$$

So,

$$E_A \perp E_B = 0$$

$$\text{And, } E_B = 4|E_C|$$

106. When the torque is deflected by a small angle θ on a dipole then,

$$\tau = pE \sin \theta$$

$$= pE\theta$$

But torque is,

$$\tau = I\alpha$$

$$pE\theta = I\alpha$$

$$\alpha = \frac{pE\theta}{I}$$

$$\text{Here, } I = (mr^2 + mr^2)$$

The small angle is given by,

$$\theta = \frac{1}{2}\alpha t^2$$

$$t = \sqrt{\frac{2\theta}{\alpha}}$$

$$t = \sqrt{\frac{2\theta}{\left(\frac{pE\theta}{I}\right)}}$$

$$t = \sqrt{\frac{2I}{PE}}$$

$$t = \sqrt{50}$$

$$t \approx 2.5\pi \text{ s}$$

107. In equilibrium condition spring force is equal to force applied between plates of capacitors.

$$\frac{q^2}{2\varepsilon_0 A} = kx$$

$$k = \frac{q^2}{2\varepsilon_0 Ax}$$

$$k = \frac{C^2 V^2}{2\varepsilon_0 Ax}$$

$$k = \frac{\frac{\varepsilon_0^2 A^2}{\left(\frac{3}{4}d\right)} V^2}{2\varepsilon_0 Ax} \quad \text{..... (1)}$$

Here, the extension in spring x is,

$$x = d - \frac{3}{4}d$$

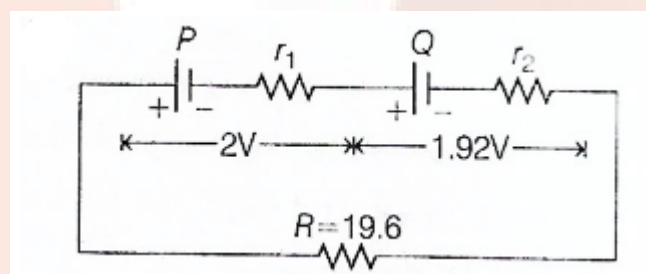
$$= \frac{1}{4}d$$

Substitute in Equation (1).

$$k = \frac{\frac{\epsilon_0^2 A^2}{\left(\frac{3}{4}d\right)} V^2}{2\epsilon_0 A \left(\frac{1}{4}d\right)}$$

$$= \frac{32}{9} \left(\frac{\epsilon_0 V^2 A}{d^3} \right)$$

108. The below figure represent the required circuit diagram.



Given information,

The Emf of each cell is equal to 2.16 V and the Emf of circuit is equal to 4.32 V.

The current in circuit is calculated by using,

$$I = \frac{4.32}{r_1 + r_2 + 19.6}$$

Use $V = E - Ir$ for each cell.

For P,

$$2 = 2.16 - \frac{4.32r_1}{r_1 + r_2 + 19.6}$$

$$\frac{4.32r_1}{r_1 + r_2 + 19.6} = 0.16 \quad \text{..... (1)}$$

For Q,

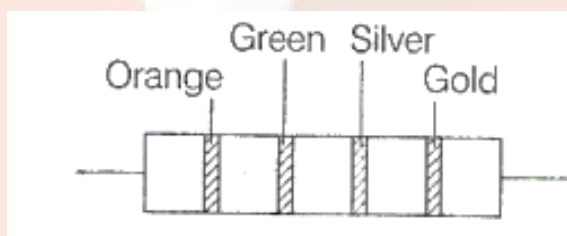
$$1.92 = 2.16 - \frac{4.32r_2}{r_1 + r_2 + 19.6}$$

$$\frac{4.32r_2}{r_1 + r_2 + 19.6} = 0.24 \quad \text{..... (2)}$$

From Equation (1) and (2) we get,

$$r_1 : r_2 = 2 : 3$$

109. The below figure represent the required diagram.



Given information:

Orange colour = 3, Green colour = 5, Silver colour = 10^{-2} and

Gold colour = $\pm 5\%$

By using colour code, the total resistance value is,

$$\begin{aligned}
 R_T &= (35 \times 10^{-2} \pm 5\%) \text{ ohms} \\
 &= (350 \times 10^{-3} \pm 5\%) \text{ ohms} \\
 &= 350 \text{ m}\Omega \pm 17.5 \text{ m}\Omega
 \end{aligned}$$

110. The path of the particle is helical, because the v is not perpendicular to B .

So,

$$\begin{aligned}
 \text{Pitch of helical path} &= v \cos \theta \left(\frac{2\pi m}{Bq} \right) \\
 &= 4 \times 10^5 \times \frac{1}{2} \times \frac{2\pi (1.6 \times 10^{-27})}{(0.314)(1.6 \times 10^{-19})} \\
 &= 4 \text{ cm}
 \end{aligned}$$

111. Magnetic field in axis is given by,

$$\begin{aligned}
 B_{\text{axis}} &= \frac{\mu N I r^2}{2(r^2 + x^2)^{3/2}} \\
 54 \times 10^{-6} &= \frac{\mu N I r^2}{2(r^2 + x^2)^{3/2}}
 \end{aligned}$$

Substitute $r = 3 \text{ cm}$ and $x = 4 \text{ cm}$.

$$54 \times 10^{-6} = \frac{\mu NI (3 \text{ cm})^2}{2 \left\{ (3 \text{ cm})^2 + (4 \text{ cm})^2 \right\}^{3/2}}$$

$$54 \times 10^{-6} = \frac{\mu NI (3 \times 10^{-2})^2}{2 \left\{ (3 \times 10^{-2})^2 + (4 \times 10^{-2})^2 \right\}^{3/2}}$$

$$\mu NI = \frac{54 \times 10^{-6} \times 2 \times (25 \times 10^{-4})^{3/2}}{(3 \times 10^{-2})^2}$$

So,

$$\begin{aligned} B_{\text{centre}} &= \frac{\mu NI}{2r} \\ &= 250 \mu\text{T} \end{aligned}$$

112. At the angle of 45° the resultant magnetic field is inclined to

B_H . So,

$$B = B_H \tan \theta$$

Here,

$$B_H = \frac{\mu_0}{4\pi} \left(\frac{M}{r^3} \right)$$

$$r^3 = \frac{M \times 10^{-7}}{B_H}$$

$$r = 10 \text{ cm}$$

113. Write the expression of length of wire in a solenoid.

$$\begin{aligned}L_w &= 2\pi r \times N \\&= 2\pi r \sqrt{\frac{Ll}{\mu_0 \pi r^2}} \\&= \sqrt{\frac{4\pi Ll}{\mu_0}}\end{aligned}$$

114. Given info:

$$I = 500 \times 10^{-3} \text{ A}$$

$$\begin{aligned}\omega_1 &= \frac{175}{\pi} \times 2\pi \text{ rad/s} \\&= 350 \text{ rad/s}\end{aligned}$$

$$V_1 = 8\sqrt{2}$$

Write the expression of current for an L – R circuit.

$$\begin{aligned}I &= \frac{V}{Z} \\&= \frac{V}{\sqrt{R^2 + L^2 \omega^2}} \\R^2 + L^2 \omega^2 &= \left(\frac{V}{I}\right)^2\end{aligned}$$

$$R^2 + L^2 (350)^2 = \left(\frac{8\sqrt{2}}{500 \times 10^{-3}} \right)^2$$

$$R^2 + L^2 (350)^2 = 512 \quad \dots\dots (1)$$

And,

$$R^2 + L^2 (550)^2 = 800 \quad \dots\dots (2)$$

By solving Equation (1) and (2) we get,

$$R = \sqrt{71} \, \Omega \text{ and } L = 60 \text{ mH}$$

115. In vacuum wavelength is given by,

$$\begin{aligned} \lambda_0 &= \frac{c}{f_0} \\ &= \frac{3 \times 10^8}{2 \times 10^6} \\ &= 1.5 \times 10^2 \\ &= 150 \text{ m} \end{aligned}$$

The speed of wave in a medium is given by,

$$\begin{aligned} v &= \frac{c}{\sqrt{\epsilon_r \mu_r}} \\ &\approx \frac{c}{\sqrt{\epsilon_r}} \end{aligned}$$

Here, the medium is non-magnetic medium.

So,

$$v = \frac{3 \times 10^8 \text{ m/s}}{\sqrt{9}}$$
$$= 1 \times 10^8 \text{ m/s}$$

Frequency remaining same, so the wavelength in the medium is,

$$\lambda = \frac{v}{f_0}$$
$$= \frac{10^8}{2 \times 10^6}$$
$$= 50 \text{ m}$$

Decreases in wavelength is,

$$\text{Decrease} = (150 \text{ m}) - (50 \text{ m})$$
$$= 100 \text{ m}$$

116. For b and a the stopping potentials are same, so

$$f_a = f_b$$

And,

The saturation current for b and a are different, thus,

$$I_a \neq I_b.$$

117. Momentum of photon for transition is given by,

$$p = \frac{h}{\lambda}$$

$$p = hR \left(\frac{1}{16} - \frac{1}{25} \right)$$

Write the expression of recoil speed.

$$v = \frac{p}{m}$$

$$= \frac{\left\{ hR \left(\frac{1}{16} - \frac{1}{25} \right) \right\}}{m}$$

$$= \frac{9hR}{400m}$$

118. The activity of a sample is given by,

$$\begin{aligned}
 R &= \left| \frac{dN}{dt} \right| \\
 &= |-\lambda N| \\
 &= \lambda N \\
 &= \left(\frac{0.6931}{T_{1/2}} \right) N \\
 &= \left(\frac{0.6931}{T_{1/2}} \right) n \times N_A \\
 &= \left(\frac{0.6931}{T_{1/2}} \right) \left(\frac{m}{M} \right) N_A
 \end{aligned}$$

By substituting the all value we get, $R = 1.26 \times 10^4 \text{ s}^{-1}$.

119. Write the expression of voltage the given circuit.

$$V_{CC} = i_B R_B + V_{BE}$$

$$R_B = \frac{V_{CC} - V_{BE}}{i_B}$$

$$R_B = \frac{V_{CC} - V_{BE}}{\frac{i_C}{\beta}}$$

$$R_B = \frac{8 - 0.6}{\frac{5 \times 10^{-3}}{50}}$$

$$R_B = \frac{8 - 0.6}{1 \times 10^{-4}}$$

$$R_B = 7.4 \times 10^4$$

$$R_B = 74 \times 10^3 \text{ k}\Omega$$

By KVL in a closed collector loop, we get.

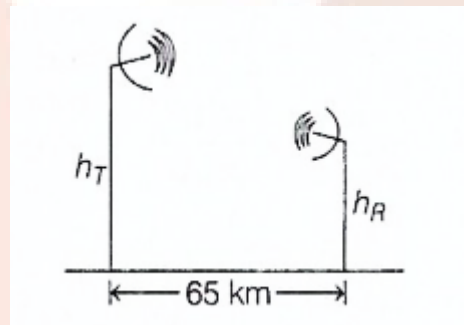
$$V_{CC} = i_C R_L + V_{CE}$$

$$R_L = \frac{V_{CC} - V_{CE}}{i_C}$$

$$R_L = \frac{8 - 3}{5 \times 10^{-3}}$$

$$R_L = 1 \text{ k}\Omega$$

120. The below figure represent the required diagram.



Here,

$$\frac{h_T}{h_R} = \frac{36}{49}$$

$$h_T = \frac{36h_R}{49} \quad \text{..... (1)}$$

The maximum diameter is given by,

$$d_{\text{maximum}} = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$65 = \sqrt{2R\left(\frac{36h_R}{49}\right)} + \sqrt{2Rh_R}$$

$$h_R = 95.7 \text{ m}$$

By substituting the value in equation we get,

$$h_T = 70.3 \text{ m}$$

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121. The wavelength (in Å) for Li^{2+} is calculated as:

$$\frac{Z_{(\text{H})}^2}{Z_{(\text{Li})}^2} = \frac{\lambda_{(\text{Li})}}{\lambda_{(\text{H})}}$$
$$\frac{(1)^2}{(3)^2} = \frac{\lambda_{(\text{Li})}}{912}$$
$$\lambda_{(\text{Li})} = \frac{1 \times 912}{9}$$
$$= 101.3 \text{ Å}$$

122. The wave number (ν) is calculated as:

$$\nu = R.Z^2 \left[\frac{1}{n_{\text{L}}^2} - \frac{1}{n_{\text{H}}^2} \right]$$

For Balmer series

$$\begin{aligned} \nu &= R \left[\frac{1}{(2)^2} - \frac{1}{(3)^2} \right] \\ &= R \left[\frac{1}{4} - \frac{1}{9} \right] \\ &= \frac{5}{36} R \end{aligned}$$

For Lyman series

$$\begin{aligned} \nu &= R \left[\frac{1}{(1)^2} - \frac{1}{(2)^2} \right] \\ &= R \left[\frac{1}{1} - \frac{1}{4} \right] \\ &= \frac{3}{4} R \end{aligned}$$

Therefore, the ratio is

$$\begin{aligned} \text{Ratio} &= \frac{5}{36} \times \frac{4}{3} \\ &= \frac{5}{27} \end{aligned}$$

123. All the given species are isoelectronic means all are having same number of electrons. For isoelectronic species, the ionic radii increase with decrease in magnitude of nuclear charge. The correct order of ionic radii is,

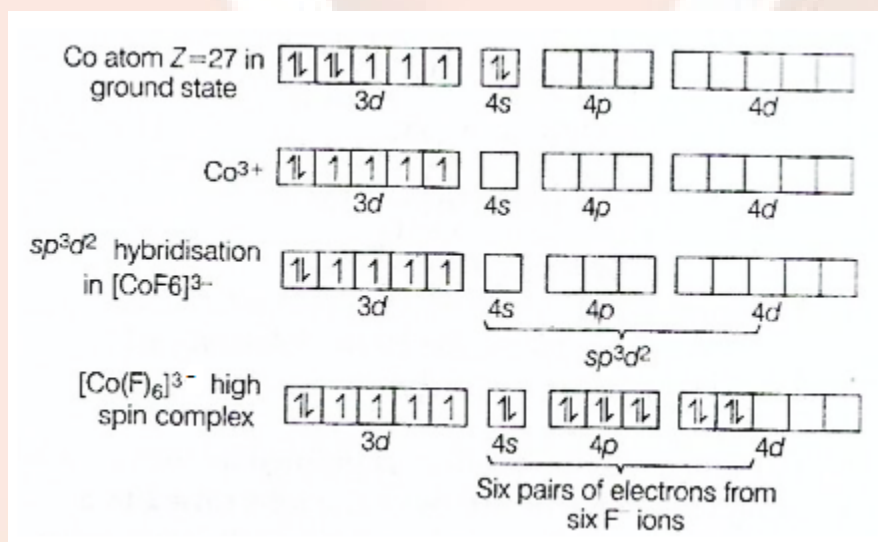
$$\text{O}^{2-} > \text{F}^{-} > \text{Na}^{+} > \text{Mg}^{2+} > \text{Al}^{3+}$$

Nuclear charge 8 9 11 12 13

124. The bond order of F_2 molecule is similar to O_2^{2-} as the bond order is equal to 1.

125. 'Co' has atomic number 27. Therefore, X is Co

The hybridization of $[\text{CoF}_6]^{3-}$ is sp^3d^2 as shown below.



126. The required temperature is calculated by using ideal gas equation as shown below.

$$pV = nRT$$

$$\begin{aligned} T &= \frac{pV}{nR} \\ &= \frac{3.32 \times 5}{4 \times 0.083} \\ &= 50 \text{ K} \end{aligned}$$

127. The total volume is $50 + 150 = 200\text{mL}$

By applying the formula as follows:

$$\begin{aligned} N_1 \times V_1 &= N_2 \times V_2 \\ 0.1 \text{ N} \times 50 &= N_2 \times 200 \\ N_2 &= \frac{0.1 \text{ N} \times 50}{200} \\ &= \frac{N}{40} \end{aligned}$$

Molarity of resultant solution is calculated as:

$$\begin{aligned} \text{Molarity} &= \frac{N}{40} \times \frac{1}{2} \\ &= \frac{M}{80} \end{aligned}$$

128. A) The change in internal energy of a system is derived by the expression $qV = \Delta U$

B) Isothermal reversible change is derived by the expression:

$$W = -p_{\text{ex}} (V_f - V_i)$$

C) Isothermal reversible change is derived by the expression:

$$W = -2.303nRT \log \frac{V_f}{V_i}$$

D) Adiabatic change is equal to $W_{\text{adiabatic}} = \Delta U$:

129. The pH is calculated as:

$$\begin{aligned} \text{pH} &= 14 - \text{pOH} \\ &= 14 - 8.6 \\ &= 5.4 \end{aligned}$$

Now, the pOH is calculated as:

$$\begin{aligned} \text{pOH} &= \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \\ \text{pOH} &= \text{pK}_b + \log \frac{[\text{NH}_4\text{Cl}]}{[\text{NH}_4\text{OH}]} \end{aligned}$$

Substitute the values

$$\text{pOH} = \text{pK}_b + \log \frac{1}{0.1}$$

$$5.4 = \text{pK}_b + \log 10$$

$$\text{pK}_b = 5.4 - 1$$

$$= 4.4$$

130. (i) Type AB



$$4.0 \times 10^{-20} = s^2$$

$$s = \sqrt{4.0 \times 10^{-20}}$$

$$= 2 \times 10^{-10}$$

(ii) Type A_2B



$$3.2 \times 10^{-11} = (2s)^2 s$$

$$3.2 \times 10^{-11} = 4s^3$$

$$s = \frac{3.2 \times 10^{-11}}{4}$$

Simplify the above equation

$$s^3 = 0.8 \times 10^{-11}$$

$$s = \sqrt[3]{0.8 \times 10^{-11}}$$

$$= 2 \times 10^{-4}$$

(iii) Type AB_3

$$K_{sp} \rightleftharpoons [A^{3+}][B^{2-}]^3$$

$$2.7 \times 10^{-31} = s(3s)^3$$

$$2.7 \times 10^{-31} = 27s^4$$

$$s^4 = \frac{2.7 \times 10^{-31}}{27}$$

Simplify the above equation

$$s^4 = 1 \times 10^{-30}$$

$$s = \sqrt[4]{1 \times 10^{-30}}$$

$$= 3.16 \times 10^{-8}$$

Therefore, the correct order is

$$AB < AB_3 < A_2B$$

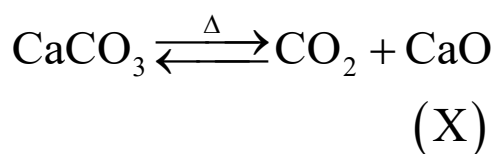
131. (i) Zn undergoes a reaction with dilute HCl and NaOH to liberate hydrogen gas.

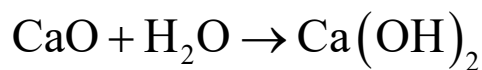
(ii) Interstitial hydrides are formed by Ti and Zr.

(iii) The viscosity of H_2O is not more than that of D_2O

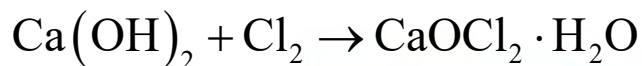
Therefore, the correct statements are (a) and (b)

132. The reactions taking place are shown below.





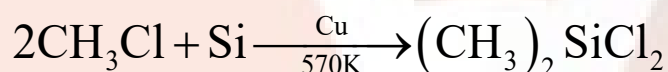
(X) (Y)



(Y) (Z)

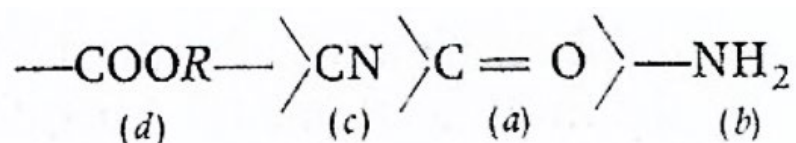
133. A molecule which can act both as an acid or base as well is known as amphoteric compound. The correct set of 13th group elements which do not form amphoteric oxides are B, In and Tl

134. The reaction is shown below.

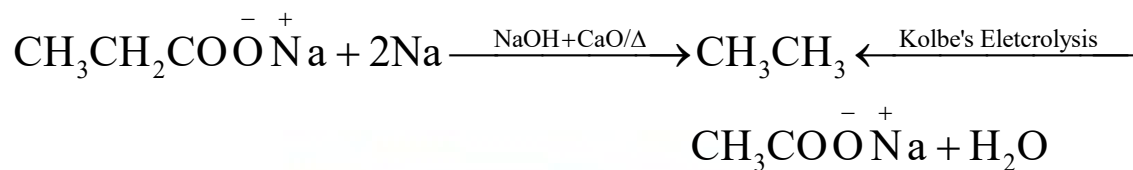


135. Nitrogen gas is not a greenhouse gas. The greenhouse gases are CO, methane, nitrous oxide and ozone.

136. For the given functional groups, the correct order of priority according to IUPAC naming is as follows:



137. The complete reaction is shown below.



Therefore,

X is $\text{NaOH} + \text{CaO}/\Delta$

Y is C_2H_6

Z is $\text{CH}_3\text{COO}^-\text{Na}^+$

138. The geometrical isomerism is not shown by prop-2-enoic acid as rotation is not restricted between carbon atoms in this molecule.

139. Octahedral voids is equal to number of atoms of metal

It is given that 3 octahedral voids fill 1 complete part of Y_3

Therefore, 1 octahedral void fills $\frac{1}{3}$ part of Y_3

The unoccupied fraction of voids is calculated as:

$$1 - \frac{1}{3} = \frac{2}{3}$$

140. The molar mass of solute (M_{solute}) can be calculated as:

$$\frac{p^{\circ} - p}{p} = \frac{w_{\text{solute}}}{M_{\text{solute}}} \times \frac{M_{\text{solvent}}}{w_{\text{solvent}}}$$

Substitute all the values

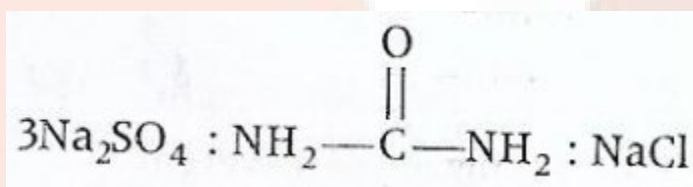
$$\frac{0.85 - 0.845}{0.845} = \frac{0.5}{M_{\text{solute}}} \times \frac{78}{39}$$

$$\frac{0.05}{0.845} = \frac{0.5}{M_{\text{solute}}} \times \frac{78}{39}$$

$$M_{\text{solute}} = \frac{0.5 \times 78 \times 0.845}{0.05 \times 39}$$

$$= 169 \text{ g/mol}$$

141. The elevation in boiling point depends upon the number of ions (Van't Hoff factor)



3 ions : 1 ion : 2 ions

Therefore, the ratio of elevation in boiling point is

3:1:2

142. Fe can displace copper and silver as its value of standard reduction potential is negative. The species which are

having negative value has tendency to displace other metals.

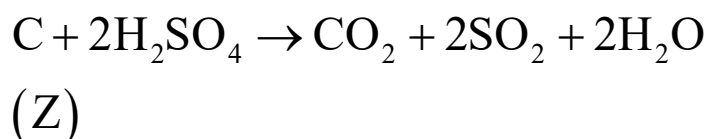
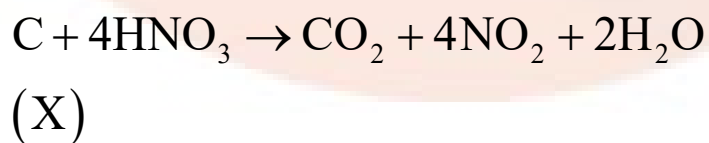
143. The time can be calculated as:

$$\begin{aligned}t &= \frac{[R_0] - [R_E]}{K} \\&= \frac{0.1 - 0.075}{2.5 \times 10^{-3}} \\&= 0.01 \times 10^{-3} \\&= 10 \text{ s}\end{aligned}$$

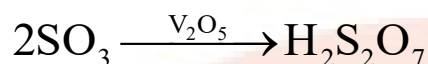
144. Kraft temperature is a temperature above which the formation of micelles does not occur.

145. Highly pure semiconductors are formed by zone refining which refines impure semiconductor metals.

146. The reactions are shown below.

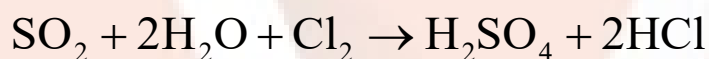


147. The catalyst such as V_2O_5 is used in contact process. The reaction of contact process is as follows:



Therefore, the correct statement is (4)

148. The reaction in which dibasic and monobasic acid is formed is shown below:



In the above reaction, H_2SO_4 is a dibasic acid.

HCl is a monobasic acid.

149. For the production of high density polymer, Ziegler Natta catalyst is used ($TiCl_4$ & $(C_2H_5)_3Al$)

150. The species Eu^{2+} and Yb^{2+} can lose electrons and produce Eu^{3+} and Yb^{3+} , therefore they are known as reducing agents.

151. The polydispersity index is calculated by the ratio of $\frac{M_w}{M_n}$

M_n is calculated as:

$$\begin{aligned}M_n &= \frac{N_1 M_1 \times N_2 M_2}{N_1 + N_2} \\&= \frac{10 \times 1 \times 10^4 \times 10 \times 1 \times 10^5}{10 + 10} \\&= 55000\end{aligned}$$

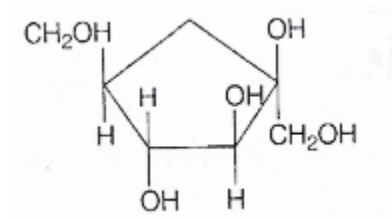
M_w is calculated as:

$$\begin{aligned}M_w &= \frac{N_1 M_1^2 \times N_2 M_2^2}{N_1 M_1 + N_2 M_2} \\&= \frac{10 \times (1 \times 10^4)^2 \times 10 \times (1 \times 10^5)^2}{(10 \times 1 \times 10^4) + (10 \times 1 \times 10^5)} \\&= 91818\end{aligned}$$

Thus, the ratio is:

$$\begin{aligned}\text{Polydispersity Index} &= \frac{M_w}{M_n} \\&= \frac{91818}{55000} \\&= 1.669 \\&\approx 1.67\end{aligned}$$

152. The structure of β - D - (–)fructofuranose is as follows



153. After attachment of chemical messenger to receptor, the shape of receptor gets changed.

Receptor cannot recover its shape when chemical messenger is removed.

Therefore, Assertion (A) is correct and Reason (R) is not correct.

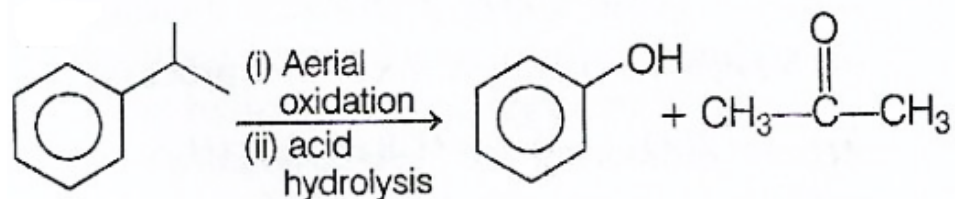
154. In structure (a), all groups are different; therefore, (a) is chiral.

In structure (b), two groups are identical; therefore, (b) is not chiral (achiral).

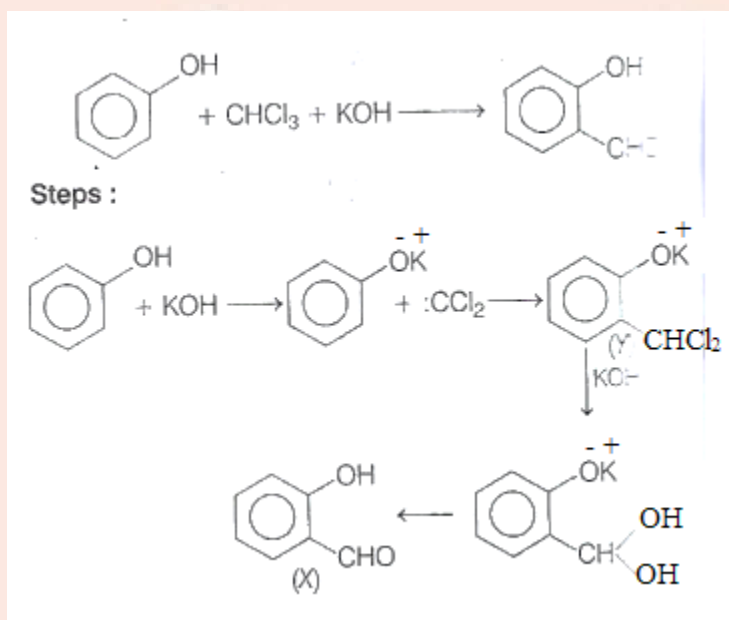
In structure (c), all groups are different; therefore, (c) is chiral.

In structure (d), two groups are identical; therefore, (d) is not chiral (achiral).

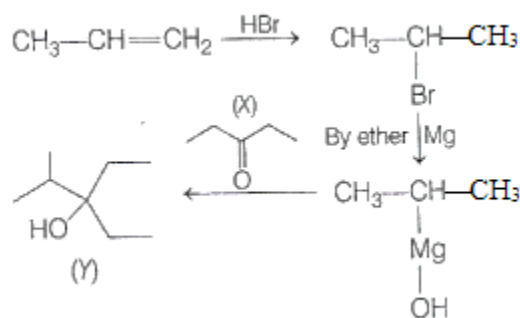
155. The resulting compounds are shown below:



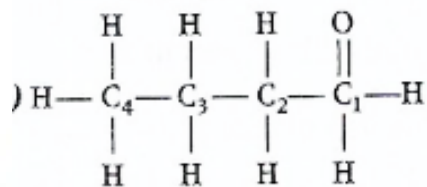
156. The products (X) and (Y) formed are shown below:



157. The products (X) and (Y) formed are shown below:



158. The hydrogen which is located at second carbon is highly acidic as it is connected with the carbonyl group.



159. $\text{C}_6\text{H}_5 - \text{COOH}$ is highly acidic as it shows $-I$ effect.
The correct order of strengths is as follows:
 $c > d > b > a$

160. The reaction is shown below.

